

Program overview

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Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Embedded Systems

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master CE/ES 2022							
Compulsory courses 2022							
CESE4000	Software Fundamentals	5					
CESE4005	Hardware Fundamentals	5					
CESE4010	Advanced Computing Systems	5					
CESE4015	Software Systems	5					
CESE4020	Effective and Responsible Engineering	5					
CESE4025	Real-time Systems	5					
CESE4030	Embedded Systems Laboratory	5					
CESE4035	Computer Arithmetic	5					
CESE4040	Processor Design Project	5					
EE4C11	Systems Engineering	5					
Specialisation courses 2022							
Specialisation courses Software (choose at least 15 EC) 2022							
CESE4075	Supercomputing for Big Data	5					
CS4200-A	Compiler Construction	5					
CS4205	Evolutionary Algorithms	5					
CS4210-A	Algorithms for Intelligent Decision Making	5					
CS4215	Quantitative Performance Evaluation for Computing Systems	5					
CS4220	Machine Learning 1	5					
CS4240	Deep Learning	5					
CS4375	Artificial Intelligence Techniques	5					
IN4049TU	Introduction to High Performance Computing	6					
IN4150	Distributed Algorithms	6					
IN4191	Security and Cryptography	5					
IN4254	Smart Phone Sensing	5					
IN4315	Software Architecture	5					
IN4344	Advanced Algorithms	5					
Specialisation courses Computer Architecture (choose at least 15 EC) 2022							
CESE4075	Supercomputing for Big Data	5					
CESE4080	Quantum Computing Architecture and Electronics - Fundamentals and state-of-the-art	5					
CESE4085	Modern Computer Architectures	5					
CESE4090	Reconfigurable Computing Design	5					
CESE4095	System Design with HDLs	2					
CS4215	Quantitative Performance Evaluation for Computing Systems	5					
EE4610	Digital IC Design	3					
EE4615	Digital IC Design II	3					
EE4690	Hardware Architectures for Artificial Intelligence	5					
EE4695	Hardware Dependability	5					
EE4700	Modelling, Algorithms and Data Structures	5					
ET4351	Digital VLSI Systems on Chip	4					
ET4362	High Speed Digital Design for Embedded Systems	5					
IN4350	Embedded Computer Architectures 2	5					
Specialisation courses Networking (choose at least 15 EC) 2022							
CESE4045	High-performance data networking	5					
CESE4050	Measuring and Simulating the Internet	5					
CESE4055	Ad-hoc Networks	5					
CESE4060	Wireless IoT and Local Area Networks	5					
CESE4065	Advanced Practical I.o.T. and Seminar	5					
CS4090	Quantum Communication and Cryptography	5					
CS4215	Quantitative Performance Evaluation for Computing Systems	5					
CS4430	Network Security	5					
EE4396	Mobile Networks	5					
EE4630	Telecommunication Network Architectures	3					
EE4C06	Networking	5					
ET4034	Telecom Business Architectures and Models	4					
ET4358	Fundamentals of Wireless Communications	5					
IN4254	Smart Phone Sensing	5					
IN4341	Performance Analysis	5					
SC42100	Networked and Distributed Control Systems	3					
Specialisation courses Control (choose at least 15 EC) 2022							
		5					

CS4215	Quantitative Performance Evaluation for Computing Systems		
RO47019	Intelligent Control Systems	4	
SC42001	Control System Design	5	
SC42015	Control Theory	6	
SC42025	Filtering & Identification	6	
SC42045	Control Systems Lab	4	
SC42056	Optimization for Systems and Control	3	
SC42061	Nonlinear Systems Theory	3	
SC42075	Modelling and Control of Hybrid Systems	3	
SC42095	Control Engineering	3	
SC42100	Networked and Distributed Control Systems	3	
SC42110	Dynamic Programming and Stochastic Control	5	
SC42125	Model Predictive Control	4	
SC42130	Fault Diagnosis and Fault Tolerant Control	4	
SC42145	Robust Control	3	
SC42150	Statistical Signal Processing	3	
SC42155	Modelling of Dynamical Systems	3	
Thesis Project 2022			
ES5000	Final Project	45	
Suggestion free elective space: Internship & Projects 2022			
EE5010	Internship 10-15 ECTS	10	
ET4399	Extra Project (max. 15EC)	15	
IFEEMCS520200	Interdisciplinary Advanced Artificial Intelligence Project	15	
TUD4040	Joint Interdisciplinary Project	15	
Language courses 2022			
TPM018A	English Grammar for the University	2	
TPM302B	Spoken English for Academic Purposes Advanced	2	
TPM303A	Intermediate Writing in English for the University	2	
TPM304A	Advanced Writing in English for the University	2	
TPM305A	Writing a Masters Thesis in English	2	
WM1115TU	Dutch Elementary 1	3	
WM1116TU	Dutch Elementary 2	3	
WM1117TU	Dutch Intermediate 1	3	
WM1135TU	Advanced English for the University	3	

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Compulsory courses 2022

CESE4000	Software Fundamentals	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. S.S. Chakraborty	
Contact Hours / Week x/x/x/x	12/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This course serves as homologation for non-computer science students that enrolled in the Computer and Embedded Systems Engineering programme. This course provides students with the fundamentals needed to successfully develop software systems in Rust, both as a single developer and when working in a team.	
Study Goals	Students will be able to: * Explain the programming language concepts followed in Rust. * Design, implement and debug a small software system from scratch in Rust following the language standard including proper coding style. * Set up a project and build environment, using the Rust ecosystem. * Use Git to version and share source code contributions for collaborative development. * Evaluate and integrate code contributions of other team members.	
Education Method	Lectures + lab	
Assessment	This is a skills-based course, so the grade will be composed of lab results (individual assignments + group project)	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for ES and CE students- they get preference over other MSc students.	

CESE4005	Hardware Fundamentals	5
Responsible Instructor	Dr.ir. J.S.S.M. Wong	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	This purpose of this course it to familiarize students with a Computer Science-focused background in topics from the field of Electrical and Control Engineering. It serves as a homologation course.	
Study Goals	The course covers 5 areas: * Basic Modeling of Systems * Analysis of Electronics Circuits * Design of Digital Systems * Basics of Computer Architecture * Design of Control Systems For each area, we have defined learning objectives. In the following, we list them per area. In the area of "Basic Modeling of Systems", after following this course, the student is able to: * describe and mathematically model simple computing systems: FSMs, transducers; * extend the modeling framework to infinite-state systems; * describe system properties and classifications; * linearize non-linear ODEs systems; and * describe and mathematically model simple physical systems: ODEs, TF, Bode plots. In the area of "Analysis of Electronics Circuits", after following this course, the student is able to: * identify different electronic (passive and active) components; * explain MOSFET characteristics; * analyze an inverter circuit and calculate noise margins; and * discuss a basic amplifier circuit. In the area of "Design of Digital Systems", after following this course, the student is able to: * distinguish between combinational and sequential circuits/modules, and use these to design simple hardware functions; * describe simple hardware functions both at the register-transfer level (RTL) and using a hardware description language such as VHDL; and * convert finite-state machines (FSMs) to hardware designs via use of combinational/sequential. In the area of "Basics of Computer Architecture", after following this course, the student is able to: * categorize different parts of a modern Computer Architecture and explain how a processor executes instructions. In the area of "Design of Control Systems", after following this course, the student is able to: * describe stability notions; * design and analyze feedback controllers for linear systems, e.g.: PID, Pole-placement, state-space design (MIMO) - LQR problem, and reference tracking; and * connect to Matlab/Python commands.	
Education Method	The course will be taught via a collection of lectures and a lab per area (5 in total - see study goals). The number of lectures is either 2 or 4 depending on the area. A detailed overview can be found on the Brightspace pages of the course.	
Assessment	The course is assessed as follows: * all labs must be successfully completed - there is a minimum threshold per lab that must be reached * there is a concluding exam (graded 1-10) for which the student must achieve a passing grade, i.e., 6 or higher.	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for ES and CE students- they get preference over other MSc students.	
Co-Instructor	Prof.dr.ir. S. Hamdioui	
Co-Instructor	Dr. R. Ishihara	
Co-Instructor	S.C. Bregman	
Co-Instructor	Ing. A.M.J. Slats	

CESE4010	Advanced Computing Systems	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Instructor	Ir. J.W. Peltenburg	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming, to have knowledge of the Linux OS, and to have followed a bachelor-level course in computer architecture. Experience in C or C++ is a plus since this course focuses on low level (closer to hardware) programming.	
Course Contents	Recent developments in computing systems have resulted in the emergence of a number of different computational platforms that provide various performance, cost and power advantages in different application domains. This course discusses the most widely used computational platforms (CPUs, GPUs, FPGAs and DSPs), while addressing the theoretical and practical trade-offs in computer system organization and the latest developments and trends in computer architecture. The course will help the students in quantifying architectural design decisions in terms of performance, cost and power. An accompanying lab aids the students in applying this knowledge to create powerful heterogeneous (CPU, GPU, FPGA and/or DSP) computational solutions in computationally intensive application domains, such as multimedia and scientific computing.	
Study Goals	After completing this course, students are able to: 1. Analyze the computational characteristics of the application (available parallelism and memory access). This helps students to understand the algorithmic limitations of running them on a multicore platform. 2. Identify the components of a multicore architecture (processing elements, memory system, interconnect network). This helps students to estimate how their application will perform on an architecture built with these components. 3. Determine the potential performance of implementing an application on various available multicore architectures. This helps students to select an optimal architecture for their application. 4. Use profiling tools to identify bottlenecks in running applications on a system. This helps students to optimize their implementation. 5. Select a specific multicore system and implement the application using various techniques such as OpenMP, CUDA/OpenCL, and SIMD extensions. This helps students to select optimal multicore platforms for their target applications.	
Education Method	Lectures for 2 hours per week in addition to lab sessions of 2 hours a week. Students are expected to spend 10 hours a week to prepare for the lab, read material and solve homework assignments at home.	
Literature and Study Materials	Publications and reading material provided via Brightspace	
Assessment	1. L1 and L2: lab assignments to use advanced computing techniques 2. L3: a lab assignment to accelerate a specific application on heterogeneous multicore platforms (CPU and GPU) 3. H: 5 homework assignments based on reading material 4. C: a multiple choice exam used to assess individual competence Final grade = $0.3 * H + C * (0.1 * L1 + 0.2 * L2 + 0.4 * L3)$ L1, L2, L3, H is in the range [0, 10] C is in the range [0, 1.1]	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Students are expected to be engaged in course activities throughout the course duration. Therefore, some students can experience a higher study intensity in this course compared to other courses in the curriculum. There will be a possibility to repeat the exam of this course. However, there is no possibility to repeat the homework or the lab. Disclaimer: the study guide information may change depending on the developments around coronavirus.	

CESE4015	Software Systems	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	This course aims to acquaint students with modern software development methods for large complex systems, in particular advanced programming language concepts (in Rust) and Model-Based Software Engineering (MBSE).	
Study Goals	At the end of the course the students will be able to: * Apply advanced programming language concepts followed in Rust, including concurrency and asynchronous execution. * Apply MBSE techniques and tools, including UML, Finite-State Machines, and Domain-Specific Languages. * Analyse the performance implications of program development at a high level of abstraction using profiling tools. * Evaluate the best approaches, ranging from zero-copy design to code refactoring, to optimize their code.	
Education Method	Lectures and labs	
Prerequisites	Literacy in Rust (CESE4000 or equivalent)	
Assessment	Graded assignments	

CESE4020	Effective and Responsible Engineering	5
Responsible Instructor	Drs. J.A. Weitenberg	
Instructor	Drs. P.C. Post	
Instructor	Dr.ir. L. Marin	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Academic Writing: The Academic Writing course teaches students to write about relevant scientific research on an academic level. They will write a literature review on a subject connected to their specialisation, in which they will review at least 15 scientific sources and learn to compare and contrast these. Students will learn to reflect on their own work and the work of their peers by doing peer reviews and processing feedback in revision rounds. Ethics: Students will familiarise themselves with the method of value-sensitive design as a method of approaching ethical problems in technological innovations. They will master this method and apply it in a case study of their choosing. Oral Presentations: The Oral Presentations course teaches students to prepare and deliver a presentation on a professional level, on a case study concerning value-sensitive design. Students will practice presenting during all lectures and will build their skills in a step-by-step approach, in which there is room for differentiation between those with less and those with further developed starting levels. In an overall reflection assignment, students will reflect on their progress in AW, OP, and Ethics, incorporating the feedback from both lecturer and peers.	
Study Goals	At the end of this course, students will: Master basic skills in academic writing. They will be able to write a literature review on a subject connected to their specialisation. Master basic principles of ethics. They will be able to recognize social and ethical issues in engineering and technology and critically evaluate possible solutions for the social and ethical issues coming up in engineering, by taking into account multiple perspectives and stakeholders. Master a professional level of presenting. They will be able to present an ethical issue connected to CESE to a mixed academic audience.	
Study Goals continuation	Academic Writing: At the end of this course, students will master basic skills in academic writing. They will be able to write a literature review on a subject connected to their specialisation. To achieve this goal they will be able to: 1. formulate a clear research question/objective 2. find, critically read and manage relevant scientific literature 3. use scientific literature properly and avoid plagiarism 4. apply the text elements according to the required genre 5. use proper scientific argumentation 6. use an academic style 7. use tables and figures to support and communicate results 8. give feedback on literature review articles 9. reflect on scientific writing and use feedback for improving their work 10. determine further learning goals for their thesis project. Ethics: Students will be able to: recognize social and ethical issues in engineering and technology and conceptualise these issues in terms of value conflicts critically evaluate possible solutions for the social and ethical issues coming up in engineering, by taking into account multiple perspectives and stakeholders. Oral Presentations: At the end of this course, students will master a professional level of presenting. They will be able to present an ethical issue connected to CESE. To achieve this goal they will be able to: 1. adjust academic and technical information to a target audience; 2. use content and rhetorical techniques to make presentations relevant and attractive to the audience; 3. use voice and body language to support the presentation; 4. apply a clear structure to a presentation; 5. use visualisation tools supporting the content of the presentation and operate these; 6. deal professionally with questions from the audience; 7. reflect on their presenting skills and formulate further personal learning goals.	
Education Method	Academic writing: 5 small-scale workshops in which students discuss and practice aspects of writing. Students will write a literature review on a subject connected to their specialization. Students will receive feedback from the lecturer as well as from their fellow students (peer feedback) to enhance their writing skills. Ethics: 2 lectures and 1 workshop on ethics. Students will then work on a case study for their final presentation. Oral Presentations: 7 small-scale workshops in which students discuss and practice aspects of presenting. Students will receive feedback on several presentation exercises from their lecturer as well as their fellow students (peer feedback) to enhance their presentation skills and prepare their final presentation on an ethical case study.	
Literature and Study Materials	Academic writing: Brightspace course TU Write Ethics: Materials will be made available on Brightspace Oral presentations: Brightspace course TU Present	
Assessment	The final grade for E&RE is based on the following components (weighing in brackets): Academic Writing: Written paper (grade) and Reflection assignment (pass/fail); (2,5 EC). Ethics: Final homework assignment (grade); (1 EC). Oral Presentations: Preparation for final presentation (pass/fail); Final presentation: 50% group presentation, 50% individual presentation techniques (grade); (1,5 EC). In case of unforeseen circumstances, or measures due to COVID-19 the final presentation will be online.	

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	- basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	The course intends to bring the student into the position to: - Explain the fundamental concepts and terminology of real-time systems - Construct task schedules using different scheduling policies under a given set of realistic system constraints - Analyze the timing behavior of a system for a given system model and scheduling policy - Discuss advantages and disadvantages of different scheduling policies for a given platform or system - Discuss the effect of hardware and software interferences on the timing behavior of a given system - Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system - Derive (reverse engineer) the system specification from a given implementation (in the lab) - Evaluate the scheduling overheads of a given implementation (in the lab) - Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Books	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	Written exam (grade) + lab work; the exam has a resit	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level.	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding@home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	Lab. project (75%) + written report (25%), no exam, no resit	

CESE4035	Computer Arithmetic	5
Responsible Instructor	Dr. S.D. Cotofana	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course emphasises on theoretical aspects of computer arithmetic. It covers concepts and various topics related to fixed and floating-point number systems, algorithms and implementations for addition, multiplication, division, square root, and other high order arithmetic operations.	
Study Goals	The study goals for the course are as follows: 1. The student can operate with concepts and notions related to: (i) number representation systems; (ii) algorithms and implementations for basic integer arithmetic operations, i.e., addition/subtraction, multiplication, and division. 2. Assuming certain processor architecture and requirements she/he can perform design space exploration and select the most appropriate algorithms for the implementation of the basic functional units. 3. She/he can operate with concepts and notions related to floating point systems and operations, elementary function evaluation, e.g., CORDIC, and error analysis. 4. She/he can design arithmetic units and application specific (co-)processors, optimized for speed, area, power consumption, or combinations of those. 5. She/he can study recent advances in computer arithmetic, classify research papers, and report about them.	
Education Method	Lectures and homework assignments.	
Literature and Study Materials	Text book: Computer Arithmetic: Algorithms and Hardware Designs, Behrooz Parhami, Oxford University Press, NY, 2000, ISBN 0-19-512583-5. Example exams with solutions available on Brightspace.	
Assessment	Exam and homework assignments contribute to the final grade as follows: written open book exam - 60% of the final grade; homework assignments - 40% of the final grade; Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	Exam is open book. Books on computer arithmetic, computer architecture, and logic design, and lecture slides are allowed during the exams.	

CESE4040	Processor Design Project	5
Responsible Instructor	Dr. S.D. Cotofana	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	To be able to complete the project the student has to have Digital Systems and VHDL knowledge. Moreover before embarking in this course she/he has to successfully complete the following courses: CESE4085 Modern Computer Architecture and CESE4035 Computer Arithmetic courses.	
Course Contents	This is an integration course that is aiming to let the students utilize computer architecture and arithmetic theoretical knowledge in the design of a general purpose processor. In this line of reasoning they have to improve the performance of a MIPS-based computing platform by focusing on computer architecture and arithmetic relevant parts of it. As starting point students receive two archives: one which includes the MIPS System on Chip VHDL code and a set of benchmarks selected from the GMPbench and MiBench suites, as well as all the necessary files required for simulation and FPGA implementation, and a second archive with the MIPS cross-compiler. To evaluate their new design, students are given a Zynq-7000 FPGA board per group.	
Study Goals	The course study goals are as follows: 1. The student can perform, assuming certain processor architecture and requirements, a design space exploration and select the most appropriate algorithms for the implementation of the basic functional units. 2. The student can integrate computer arithmetic and computer architecture knowledge and utilize them in order to design/optimize processors. 3. The student is able to evaluate processor performance based on benchmarks and take the appropriate architectural decisions in order to improve on various design metrics. 4. The student can design arithmetic units and processors and implement them using FPGA technology. 5. The student is able to operate in a small team and collaborate for the successful completion of a design assignment. 6. The student can communicate his/her proposal, experimental results, and conclusions in English using the appropriate technical language in written as well as orally.	
Education Method	Lectures and Project.	
Literature and Study Materials	There is no dedicated study material but the students are advised however to utilize as supporting study material Computer Arithmetic: Algorithms and Hardware Designs, Behrooz Parhami, Oxford University Press, NY, 2000, and Computer Architecture: A Quantitative Approach, J.L. Hennessy and D.A. Patterson, 4th edition. Moreover we encourage you to utilize the IEEE Explore library to find research publications that may help you in the successful project completion.	
Assessment	To complete the course students have to submit the project archive of the new core they developed, a report describing their approach and findings, and to give a short presentation of their project in the context of a symposium. The new core functionality is verified and all projects are checked for (between groups) plagiarism. Non functional designs do not pass the project. Plagiarism can also make you fail. Your final score for the project is determined based on the following criteria: The performance of your design (DP). The benchmark scores you report are important. The higher the benchmark score you get the better but this is not the only relevant aspect. In the performance evaluation we also take into consideration the other metrics with more emphasis on the compound ones. The technical merit of your approach (TM). Aspects as innovation level and implementation quality are considered. The report (R). Report organization, content, and language are important aspects at this point. The presentation (P). Here we also look at the capability to ask questions and to answer questions from the auditorium. The ET4 171 final grade is computed as: $\text{Grade} = 0.35 \cdot \text{DP} + 0.35 \cdot \text{TM} + 0.20 \cdot \text{R} + 0.10 \cdot \text{P} - \text{C},$ where C can assume values between 0 and 1 and reflects: (i) the lack of collaboration in the group; (ii) the amount of consultancy you asked me for during the project completion. If the group is functioning like a team and you do not ask too much help C is 0, otherwise it can get values up to 1. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

EE4C11	Systems Engineering	5
Responsible Instructor	P.P. Sundaramoorthy	
Responsible Instructor	Dr. A. Endo	
Responsible Instructor	S. Izadkhast	
Instructor	S. Izadkhast	
Instructor	Dr.ir. R. Vismara	
Instructor	Dr. F. Fioranelli	
Instructor	C. Galuzzi	
Instructor	Prof.dr. P.J. French	
Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr.ir. C.J.M. Verhoeven	
Instructor	Prof.dr.ir. R.E. Kooij	
Instructor	R.T. Rajan	
Instructor	P.P. Sundaramoorthy	
Contact Hours / Week x/x/x/x	7/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Problem Formulation - Requirements specification - Use case model - Functional Decomposition - Means and Morphological Analysis - Decision Making - Simulation and modeling - Risk Analysis - Testing	
Study Goals	By the end of this course, you will be able to: - Recognize the need and relevance of systems engineering (SE) - Structure the key steps in Systems/Product Development - Systematically breakdown tasks and work in teams towards a common goal - Apply SE methodologies and SE tools in projects	
Education Method	he course is centered around 5 group project sessions. You will work in a group of about 10 students with different backgrounds. Each session focuses on one or a few steps in Systems Engineering. The course will be organized in 3 phases. At the end of phase 1 and phase 2, there will be a midterm review. At the end of the course there will be a final review, and a symposium.	
Assessment	The final grade will be composed of: - Midterm review (x2) - Final review - Final report - Individual assessment	
In addition, there will be formative assessment (i.e., not graded) in the form of feedback from the TA and Tutor, and peers.		

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Specialisation courses 2022

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Specialisation courses Software (choose at least 15 EC) 2022

CESE4075	Supercomputing for Big Data	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming. Experience in Java, Scala or Python is a plus due to their importance in the big data ecosystem.	
Course Contents	Big data is one of the hottest technology fields today, used to describe large and complex data sets that are difficult to process using traditional data processing systems. In this course, we will introduce the students to the most important concepts of big data analytics and the available tools and systems used to manage and process it. In a series of labs, the students will be working with a number of different big data applications, addressing such aspects as implementing big data algorithms using Spark and Hadoop on high performance computational systems, as well as using higher-level languages and system management tools to guide and monitor the performance of these systems. The students will have access to a big data cluster to evaluate the effectiveness of their implemented solution in practice.	
Study Goals	By the end of this course, the students will be able to - use basic big data processing systems like Hadoop - implement parallel algorithms using the in-memory Spark framework, and streaming using Kafka - use libraries to simplify implementing more complex algorithms - identify the relevant characteristics of a given computational platform to solve big data problems - utilize knowledge of hardware and software tools to produce an efficient implementation of the application	
Education Method	Lab assignments and reading material provided via Brightspace and online on the CognitiveClass.ai site	
Assessment	The students will be evaluated based on the quality of their implementation and the report they submit describing their solution approach. The course has three lab assignments: L1, L2 and L3. The final grade of the course will be evaluated as $\text{Grade} = 0.5 * L1 + 0.3 * L2 + 0.2 * L3$ Disclaimer: information may change depending on the developments around coronavirus.	

CS4200-A	Compiler Construction	5
Responsible Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	- programming (required) - software engineering (recommended) - concepts of programming languages (recommended) - formal languages and automata (recommended)	
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assembly code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization	
Study Goals	At the end of this course, you will be able to: - Explain the architecture of a compiler and the role of each component of that architecture - Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples - Implement a code generator that translates source language abstract syntax trees to target language instructions - Implement transformations on abstract syntax terms to simplify programs - Implement simple type checkers - Implement simple data-flow analyzers - Integrate these components into a working compiler	
Education Method	Attending the course involves: - Attending weekly lectures - Attending weekly lab session (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)	
Literature and Study Materials	Required reading: - We will use selected chapters from the book "Essentials of Compilation" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation - Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.	
Assessment	The final grade will be determined by the following components - final exam (50%) - homework assignments (50%)	

	Separate grades are given to each of these components, but only the final grade will be registered in Osiris. The grades for each of the components should be at least 5.0 and the final grade should at least be 5.8. The final exam is a written 3-hour exam. Grades of the project or written exam do not automatically carry over from previous years, so upon retaking the course talk to your lecturer first. Disclaimer: information may change depending on the developments around the coronavirus.
Permitted Materials during Tests	No materials are permitted during the exam.
Judgement	to be decided
Co-Instructor	Dr. S.S. Chakraborty

CS4205	Evolutionary Algorithms	5
Responsible Instructor	P.A.N. Bosman	
Instructor	Dr. M. Virgolin	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Must have the ability to program, predominantly in Python (but also C/C++ can be useful) so that you can create your own (extensions of) evolutionary algorithms (e.g., C, C++, Java). Further, although no particular high-level prior knowledge is expected, a basic understanding of key aspects in linear algebra (e.g., matrix multiplications, inversions, decompositions, etc.), probability theory and statistics (e.g., normal distribution, statistical hypothesis testing), algorithm design (e.g., efficient local search, heuristics), complexity theory (e.g., big-O notation, algorithm analysis), will be very helpful.	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, and real-world applications. The course is planned to have 8 lectures, multiple small practical assignments and 1 larger practical assignment. The first practical assignments pertain to experimenting with already implemented EAs on predefined problems. The second practical assignment offers more freedom, allowing you, in a group, to build your own EA (this may vary depending on student numbers and other circumstances).	
Study Goals	Upon successful completion of this course, students will be able to: 1) Explain the key concepts underlying the main streams in Evolutionary Algorithm (EA) research, with in particular genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2) Explain key ingredients underlying the rationale of when these algorithms work and when they do not work. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3) Name and explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4) Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5) Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	8 Lectures 4 smaller, individual, lab projects 1 larger, lab project in groups (of 4)	
Literature and Study Materials	Papers and slides that will be made available.	
Assessment	The final grade is based on 60% written exam, 40% lab practical work. disclaimer: information may change depending on the developments around the coronavirus. In particular, there may be no written exam. In that case, there will likely be several smaller practical assignments and a large practical assignment at the end of the course. Both parts will then likely count for 50%. For the large assignment there will also be chance to resit this part by means of a repair option through an oral examination of the lecture contents.	
Permitted Materials during Tests	None	
Tags	Algorithmics Artificial intelligence Optimisation	

CS4210-A	Algorithms for Intelligent Decision Making	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	Dr. E. Demirovi	
Instructor	S. Dumanı	
Instructor	Dr. V. Robu	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: IN4010: Artificial Intelligence Techniques, or equivalent; and/or IN4301: Advanced Algorithms, or equivalent Required: basic course(s) in algorithm design and analysis, logic and probability; basic programming (in Python)	
Course Contents	Decision making is at the centre of artificial intelligence. This course gives you practical skills on a solid theoretical base. The course looks at solving mathematical models of NP-hard discrete optimisation problems. These kinds of problems lie at the heart of AI techniques such as planning, machine learning and mechanism design, and more generally combinatorial optimisation. You will learn about a range of modelling techniques from boolean satisfiability to constraint programming, and how advanced solvers for these models work. The course has plenty of real-world case studies as well as theoretical results.	
Study Goals	Apply the skills you learn in this course by taking CS4210-B: Intelligent Decision Making Project in quarter 4! By the end of this course, you will be able to identify features of real-world combinatorial decision problems, and be able to model and design systems for simplified instances of these problems using boolean satisfiability, mixed integer programming, and constraint programming over finite and real domains. You will be able to explain how SAT, CP and LCG solvers work in some detail, and how MIP solvers work at a high level. You will also be able to explain the basics of program synthesis and combinatorial game theory.	
Education Method	Lectures, homework exercises (optional), and programming assignments. The expected workload is: 30% lectures (including preparation for the exams) 30% homework exercises (optional) 40% programming assignments	
Literature and Study Materials	Provided on Brightspace	
Assessment	The final grade depends on the grades obtained for (a) programming assignments (1 or 2 in total) [40%] and (b) the exam [60%]. The final grade is computed from the unrounded grades for the components. The final grade for the programming assignment is a uniformly-weighted average of the unrounded grades obtained for the two assignments. Programming assignments can be completed by two students working together. The exam is graded on a scale from 1 to 10. A resit will be available for the exam. The result for the exam is determined by the maximum score obtained for the original exam and the resit. In order to pass the course, the rounded grade (after resit if applicable) for each part of the course must be at least 5.0, and the rounded final grade on the course must be at least 5.8. disclaimer: information may change depending on the developments around the coronavirus, and the amount of teaching support.	
Elective	Yes	
Tags	Algorithmics Artificial intelligence Group work Modelling Optimalisation Programming Projects Small groups	

CS4215	Quantitative Performance Evaluation for Computing Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Today's computing systems become ever complex, due to the rapid development of hardware and software technology. It is challenging to design and run computing systems that guarantee users performance requirements in a resource efficient way. Various quantitative methods are applied to capture such complex system dynamics and predict metrics of interests, from the designing phase of the systems to the runtime performance, e.g., job response times and system anomaly. To optimize the performance of computing systems, a deep understanding on those methods and their applications on the system design are essential. Having practical hand-on experience on designing experiments, deriving models, and validating results with benchmark systems will prepare students to tackle challenges of real world computing systems.	
Study Goals	LO1. Design full/fractional factorial experiments for multi-variate regression analysis, e.g., finding critical parameters for deep learning clusters LO2. Apply queueing theory to analyse and predict run-time performance of applications, e.g., the average response times of on-line ML training service LO3. Apply machine learning models to analyse and predict the system dependability, e.g., root cause analysis for machine failure. LO4. Conduct experiments to profile applications and extract their workload parameters on real systems, e.g., e.g., deep learning clusters LO5. Develop resource management policies and validate them on real computing systems, e.g., deep learning clusters	
Education Method	Lectures: 7 weeks X 2-4h Practical: Derive, validate and evaluate performance models and resource management strategies for a chosen system via homework and group project. Multiple types of computing and network systems can be chosen from. Deliverables include a report and group presentation.	
Books	1) Performance Modeling and Design of Computer Systems: Queuing Theory in Action by Mor. Harchol-Balter 2) Design and Analysis of Experiments by Douglas C. Montgomery, 3) The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer Series in Statistics.	
Assessment	Homework (40%): 2 individual homework Group project (60%): group project report and presentation	
disclaimer: information may change depending on the developments around the coronavirus.		

CS4220	Machine Learning 1	5
Responsible Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/4/0/0 Lectures + 0/2/0/0 lab	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	This course is required for CS4230 Machine Learning 2	
Expected prior knowledge	For the course CS4220, you should know the terminology that is taught in the course CSE2510. So, please have a look at the content of CSE2510 in Brightspace. It is not required that you followed the course CSE2510 in full, or made the exam.	
Course Contents	Recapitulation of (un)supervised learning, classification, decision theory overfitting. Complexity, regularisation, and support vector classifiers. Regression, linear and kernel regression. Bayesian learning, graphical models. Clustering and mixture models, the EM algorithm. Feature selection and extraction, PCA. Design and analysis of ML experiments.	
Study Goals	After successfully completing this course, the student is able to: recognise machine learning problems and select algorithms to solve them; read and comprehend recent articles in engineering-oriented pattern recognition journals, such as IEEE Tr. on PAMI; construct a learning system to solve a given simple machine learning problem, and able to implement algorithms from literature.	
Education Method	Lectures, laboratory work (mathematical exercises and computer exercises)	
Assessment	One final exam for 100% of the grade. This can be a written exam or an online exam, depending on the Corona situation.	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
	M. Loog	

CS4240	Deep Learning	5
Responsible Instructor	Dr. J.C. van Gemert	
Instructor	Dr. D.M.J. Tax	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	Seminar Computer Vision by Deep Learning	
Expected prior knowledge	Basic pattern Recognition, Basic Machine learning, Basic statistics, Basic probability theory, Programming experience (python + numpy).	
Course Contents	Note: Without some affinity/experience with python (and numpy) programming it might still be possible to pass the course, but will become quite difficult. In this course we will look at a specific field of Artificial Intelligence and Machine Learning: Deep learning. Deep learning has shown remarkable success with large data sets and unstructured input data such as raw images/audio/text. Topics include: feed forward networks, back-propagation, optimization, convolutional nets, recurrent nets, self-attention, unsupervised methods. The course will have lectures, a seminar, a lab practical and a project: - The lectures will be on generic topics, following the book; building the backbone. - The lab assignments will have you apply basic concepts of the lecture in python notebooks. - The seminar will have students read, critique, and present relevant deep learning research papers. You will have to read 2 papers per week, for 7 weeks (14 papers). - The project will have students apply and design their own (small) deep learning project in the context of scientific reproduction.	
Study Goals	Upon successful completion of the course, students will be able to: [LO1]. Describe the different deep learning techniques reviewed in the course, such as SGD, MLPs, CNNs, RNNs, GANs. [LO2]. Research literature concerning one of the above techniques, summarize it and report it to your peers [LO3]. Debate upon positive and negative aspects of techniques and research papers [LO4]. Quickly identify the core contributions of a research paper [LO5]. Implement one or more of the above mentioned techniques in a computer language and deep learning toolkit (eg: Pytorch) [LO6]. Determine which technique(s) is most appropriate for being used in a certain problem domain. [LO7]. Apply the appropriate technique to a (simple) problem domain which may need to re-implement, run, evaluate, investigate, extend existing research or code [LO8] Write clearly and concisely about your code, method, results, and analysis.	
Education Method	Lectures for basic theory based on the literature Assignments: we help you to become familiar with PyTorch; applying concepts from the lecture on small problems. Lab project: design and execute your own deep learning project in the context of scientific reproductions using https://reproducedpapers.org/ Seminar: paper reading, critiquing, and presenting.	
Literature and Study Materials	Books: freely available online: - http://www.deeplearningbook.org/ - https://d2l.ai/ Research papers will be made available through Brightspace.	
Assessment	Assignments are based on PyTorch: https://pytorch.org/ 1. 0.1 weight: Presentation: during the seminar a small group of students presents a paper. You will have to present once. 2. 0.9 weight: Exam about the papers, assignments, and the theory. 2. Pass/Fail: Project: in a small group of students you work on a deep learning paper reproducibility project. Final grade: (pass project) * (0.1 * Presentation + 0.9 * Exam) *** Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) ***	

CS4375	Artificial Intelligence Techniques	5
Responsible Instructor	Dr. F.A. Oliehoek	
Instructor	Dr. M.T.J. Spaan	
Instructor	Prof.dr. C.M. Jonker	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	We expect students to have programming expertise at the Bachelor level of Computer Science, in particular knowledge is expected of algorithms (e.g. search algorithms), and probability theory. Programming skills are required for the practical assignments which can be done in Python.	
Course Contents	The course focuses on concepts and techniques from probabilistic Artificial Intelligence for building intelligent agents, and decision making and -support systems. The specific topics include Bayesian networks, inference, belief monitoring, (partially observable) Markov decision processes, reinforcement learning and multiagent systems.	
Study Goals	After completing the course, student will be able 1. to comprehensively summarize the covered topics within probabilistic artificial intelligence and discuss their main issues; 2. to explain the main concepts and techniques; 3. to relate practical problems to appropriate AI techniques and apply the formal models covered in the course to these problems; 4. to implement and evaluate algorithms for complex decision making.	
Education Method	Lectures, lab work (programming assignments).	
Literature and Study Materials	Stuart J. Russel and Peter Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Prentice-Hall. ISBN 9781292401133 Global edition. See website http://aima.cs.berkeley.edu/global-index.html for additional information that goes with the book.	
Assessment	Written exam and practical assignments. The grade of the course is determined by the grade of the written exam (80%) and the mean grade of the practical assignments (20%). There are no resit possibilities for the practical assignments. There will be a resit possibility for the written exam.	
Remarks	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) The practical assignments will be done in groups of 4 to 5 students. The ethical standards of working are expected of all students. Work can be divided over students, but all students are responsible for the overall quality and originality of the work. Students that do not do their share of the work have to be reported by the group to the supervisor so that action can be taken.	
Tags	Artificial intelligence	

IN4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations; some experience with a programming language (e.g., C) is preferred but not required.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas is considered. The scaling behaviour of these algorithms in case of an increasing problem size and/or an increasing number of processors, is analysed. Attention is paid to those aspects of computer architectures that are important to understand the resulting performance, such as the memory hierarchy and the interconnection network. By analysing a number of case studies (applications) with respect to their computing-intensive character, possible bottlenecks will be determined. Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to get a high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and the necessary optimisation for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found in the Brightspace.	
Course Contents Continuation	High Performance Computing, parallel programming, parallel algorithm	
Study Goals	1. Knowledge about high performance computer systems including parallel and distributed architectures, and programming models; 2. Basic knowledge about the concepts of data decomposition and parallel algorithms; 3. Knowledge about various high performance (numerical) methods and their parallelization; 4. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 5. Obtain some experience on performance analysis of parallel programs.	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): implementing (small) parallel programs with C, MPI and Cuda.	
Literature and Study Materials	Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	Written exam (50%) + Lab work (50%).	
	disclaimer: information may change depending on the developments around the coronavirus.	
Enrolment / Application	Via Brightspace	

IN4150	Distributed Algorithms	6
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	Understand the main fundamental problems in distributed systems Understand the most important distributed algorithms that solve these problems Be able to reason about the execution of distributed algorithms Be able to program distributed algorithms Be able to select and summarize relevant literature on distributed algorithms	
Education Method	Lectures and lab work executed in groups of two students	
Literature and Study Materials	Lecture notes and lecture slides (available on Brightspace)	
Assessment	One paper summary and an in-person, on-campus written exam (closed book), which together are an integrated whole. The grade of this exam, which includes the grade of the paper summary (with a weight of 25%), is the grade for the course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) In case in-person, on-campus written exams are not possible because of unforeseen circumstances or measures, the written exam will be replaced by individual online oral exams.	
Permitted Materials during Tests	None except the list of algorithms	
Remarks	Lab work is 40 hrs.	

IN4191	Security and Cryptography	5
Responsible Instructor	Z. Erkin	
Instructor	Dr. K. Liang	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	UT-201500042 Privacy Enhancing Technologies (Q4)	
Expected prior knowledge	Basic understanding on the following is suggested. -Probability and statistics -Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Learning outcomes: The goal is to make students familiar with the basic concepts applied cryptography, including classical cryptography and modern secret key and public key cryptography. In particular, the students will acquire A sound understanding of the notion of security An understanding of the confidentiality, integrity and authenticity needs of the society Understand the role of cryptographic primitives including the differences between symmetric and asymmetric cryptography, the role of hash functions, digital signatures and PKI Understand the advanced topics in cryptography needed for the modern society with untrustworthy entities Among others things, the following topics are covered: -Classical systems -Information theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures, -*Secret Sharing. (if time permits) -*Zero Knowledge proofs (if time permits)	
Study Goals	It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security, as well as is familiar with present applications.	
Education Method	Through assignments, students are expected to have the chance to work on the topics covered in the lectures. Lectures, assignments and weekly exercises. Attention: This course requires full effort of 140 hours. Even more, if you lack the background (probability and modular arithmetic) Planned Workload: Lectures: 28 x 45minutes sessions, total 22 hours Practice session: 7 x 90 min. total 12 hours Assignments: 3 x 20 hour, total 60 hours Weekly study: 7 x 4 hours, total 28 hours Exam preparation: 20 hours Exam: 3 hours	
Literature and Study Materials	Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online)	
Assessment	Handouts of lectures Written exam(70%) + mandatory assignments (30%). Passing grade for the written exam and assignments is required. Please refer to CS regulations for further details. There is NO reparation for assignments. Points from previous years cannot be transferred.	
Exam Hours	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) In case of in person examination at campus: The exam is closed book. A cheat sheet of size A4, hand written is allowed for the written exam. Name and student number has to be present on each side. In case of remote exam from home: open book: textbook, slides and self-made notes only. randomised/customised exam.	

Permitted Materials during Tests	Only non-scientific calculators.
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IN4254	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course TI 1206 in our first year Bachelor curriculum "Object Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course TI 2216M in our second year Bachelor curriculum "Probability and Statistics".	
Course Contents	We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming. The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved on research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve the particular problems in their areas. After taking this course students will be able to: (1) Explain the current applications, methods and research trends in the area of smartphone sensing. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non-trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	Written reports + project presentation + oral exam Overall, the final grade is determined by: 1) Two intermediate reports (5% of grade each, 2 pages each) 2) Final report (10 % of grade, 5 pages) 3) Final project demonstration (80% of grade) The first two reports are due on the third and fifth week; and the final report, project and exam are due on the ninth week. There is no resit for this course.	
Enrolment / Application	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) 1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with ES students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot. IMPORTANT: The study guide information may change depending on the developments around the coronavirus.	

IN4315	Software Architecture	5
Responsible Instructor	Prof.dr. A. van Deursen	
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Software engineering	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four, in which the actual architectures of existing open source systems are analyzed in considerable detail. These systems are taken from github, and student teams are challenged to actually contribute to the systems under analysis in the course.	
Study Goals	Bring students into the position that they can (1) explain the key architectural concepts and methods for modeling software architectures; (2) apply viewpoints and perspectives to model software architectures; (3) discuss the benefits of architecting and the role of the software architect; (4) evaluate and validate software architectures; (5) explain and discuss the concepts of component-based and plugin architectures, service-oriented architectures, and software product lines; (6) explain and recognize technical debt and have an understanding of possible refactorings.	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books; Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; and Coplien and Bjørnvig, "Lean Architecture", Wiley, 2010. Additional reading material will be announced in the lectures.	
Assessment	No written exams. Grades will be based on lab assignment including essay writing, coding, (video) presentation, peer reviewing, participation. Disclaimer: information may change depending on the developments around the coronavirus.	
Special Information	Course web site: https://se.ewi.tudelft.nl/delftswa/index.html	
Co-Instructor	M. Finavaro Aniche	

IN4344	Advanced Algorithms	5
Responsible Instructor	Dr. M.M. de Weerd	
Instructor	Prof.dr.ir. K.I. Aardal	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic courses in Algorithmics and Complexity Theory	
Course Contents	The course is on solving (abstract models of) complex real-world problems, with a focus on solving intractable problems exactly. The course consists of two main parts: * modeling and solving using linear programming * exact algorithms using search trees, dynamic programming, and/or decision diagrams	
Study Goals	By the end of this course, students will have knowledge of and experience with the following advanced algorithmic techniques: (Part 1:) - linear programming (LP) and LP modelling - duality and simplex algorithm - integer LP and integer modelling - LP relaxation and branch and bound (Part 2:) - complete and bounded search trees - tree decomposition - dynamic programming - preprocessing - decision diagrams Furthermore, they will be able to * recognize situations where they can use these techniques and adapt them to different applications. - construct new algorithms that are similar to these techniques. - construct proofs that are similar to a selected set of proofs from the material. - analyze the run time of algorithms. - implement an algorithm that is given in pseudo-code. - experimentally evaluate the quality and the runtime of an algorithm on a set of instances.	
Education Method	Interactive lectures, optional homework exercises, programming assignments The expected workload is 30% studying the written material and the recorded lectures, and participating in the interactive sessions 30% making the homework exercises and preparation for the exams 40% working on the programming assignments	
Literature and Study Materials	Part 1 of the course will be mainly based on chapters 1-8, some of 9-11 (most of 9-11 is assumed to be known), and 12-14 of the syllabus "Optimization" by Karen Aardal, Leo van Iersel and Remie Janssen, which can be ordered via https://www.webedu.nl/bestellen/tudelft/ Part 2 will use chapter 10 of the following textbook: J. Kleinberg and E. Tardos, Algorithm Design, Pearson Education, 2006. ISBN 0-321-37291-3	
Assessment	As well as some supplemental study material, which will be provided via BrightSpace. The final mark depends on the marks obtained for (a) written reports and evaluation of programming assignments, PA (weight 40%) (b) a written exam, EX (2 parts, weight 60%) Each programming assignment is graded on a scale from 0 to 10. The final mark for the programming assignment (PA) is the average of the mark obtained for the assignments. Programming exercises can be completed by 2 students working together. The exam consists of two parts. Each part will be examined after the lectures about that part have been delivered and will be graded on a scale from 1 to 10. The final mark for the exam (EX) is the average of the marks for the parts. Each exam part contains one or more challenging assignments; during each part of the course two representative homework assignments will be made available to prepare for this. There is a resit for the exam where any of the two parts can be redone. The result for a part after the resits is determined by the maximum score obtained for the original exam and the resit. There is no repair option for the programming assignment. The final mark for the course is determined as follows: - if the PA and the EX mark are above 5, the final mark is the weighted average of these three marks: 60% EX, 40% PA - if at least one of PA, EX is less than or equal to 5, the final mark is the minimum of the results obtained for PA or EX. Partial results are valid only in the current academic year.	
Special Information	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) For content questions / of general interest, please use StackOverflow. For personal questions, please use the course email address: aa-cs-ewi@tudelft.nl	
Tags	Algorithmics Artificial intelligence Mathematics	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Embedded Systems

Specialisation courses Computer Architecture (choose at least 15 EC) 2022
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CESE4075	Supercomputing for Big Data	5
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming. Experience in Java, Scala or Python is a plus due to their importance in the big data ecosystem.	
Course Contents	Big data is one of the hottest technology fields today, used to describe large and complex data sets that are difficult to process using traditional data processing systems. In this course, we will introduce the students to the most important concepts of big data analytics and the available tools and systems used to manage and process it. In a series of labs, the students will be working with a number of different big data applications, addressing such aspects as implementing big data algorithms using Spark and Hadoop on high performance computational systems, as well as using higher-level languages and system management tools to guide and monitor the performance of these systems. The students will have access to a big data cluster to evaluate the effectiveness of their implemented solution in practice.	
Study Goals	By the end of this course, the students will be able to - use basic big data processing systems like Hadoop - implement parallel algorithms using the in-memory Spark framework, and streaming using Kafka - use libraries to simplify implementing more complex algorithms - identify the relevant characteristics of a given computational platform to solve big data problems - utilize knowledge of hardware and software tools to produce an efficient implementation of the application	
Education Method	Lab assignments and reading material provided via Brightspace and online on the CognitiveClass.ai site	
Assessment	The students will be evaluated based on the quality of their implementation and the report they submit describing their solution approach. The course has three lab assignments: L1, L2 and L3. The final grade of the course will be evaluated as Grade = 0.5 * L1 + 0.3 * L2 + 0.2 * L3 Disclaimer: information may change depending on the developments around coronavirus.	

CESE4080	Quantum Computing Architecture and Electronics - Fundamentals and state-of-the-art	5
Responsible Instructor	F. Sebastiano	
Responsible Instructor	Dr. S. Feld	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students should have successfully completed the course Fundamentals of Quantum Information (AP3421). Basic knowledge of Matlab is also required.	
Course Contents	The goal of this course is to introduce the students the overall system of a quantum computer, focusing on the classical hardware and software infrastructure required to build a quantum computer together with the quantum hardware. This course complements the course Quantum Hardware (AP3292) that covers the quantum components in a quantum computer by covering all classical hardware and software components. Covered topics: - Quantum computer architecture; quantum languages; compilers; QISA; microarchitecture. - Quantum simulator (QX simulator); - Quantum error correction: quantum error correction codes; encoding; logical operations. - Mapping of quantum circuits; placement of qubits; scheduling of operations; routing of qubits; - Quantum Classical Hybrid Computation; - Classical hardware for quantum computing: classical controller and quantum processor; qubit hardware and the need for cryogenic electronics; - Receiver and transmitter architectures for controlling qubits: frequency up/down-conversion, modulation schemes, frequency generation. - Hardware building blocks: amplifiers, analog-to-digital converters, digital-to-analog converters, digital processing, FPGA - Cryo-CMOS: device characteristics, state-of-the-art, design principles.	
Study Goals	By the end of this course, students are able to: - Explain the different layers that are required to build a quantum computer. - Discuss the basic functioning of quantum error correction (QEC). - Carry out the mapping step of simple quantum circuits onto specific physical device layouts. - Interpret the results of different quantum circuits written in language QASM and executed using simulator QX, including the following: Basic quantum algorithms; Fundamental QEC codes; Surface code, especially for a Ninja star. - Identify and describe the different hardware blocks in the classical controller in a quantum computer. - Derive the specifications of the individual components in a classical controller from the requirements of the whole controller. - Compare different hardware blocks based on their specifications and can select the appropriate component for target functionality and specifications. - Describe the main features and advantages/disadvantages of the adoption of cryogenic electronics in general and cryo-CMOS specifically.	
Education Method	Lectures.	
Assessment	Weekly homework and a final assignment. Final grade computed as follows: 50% homework; 50% final assignment.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

CESE4085	Modern Computer Architectures	5
Responsible Instructor	Dr. S.D. Cotofana	
Instructor	Dr.ir. J.S.S.M. Wong	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	The course emphasises on theoretical aspects and practical trade-offs of computer architecture and organization of the newest microprocessors currently on the market, and the latest developments in computer architecture research. Quantifying design decisions in terms of performance and cost. Subjects: basic principles, instruction set architectures, pipelining and pipelining consequences, multiple-issue (superscalar and VLIW) processors, multimedia SIMD extensions, out-of-order execution, branch prediction, speculative execution, advanced memory hierarchies, pre-fetching, multithreaded processors and multiprocessors, energy consumption, and reliability.	
Study Goals	1. The student can operate with concepts and notions related to: - Instruction sets: characteristics, functions, formats, addressing modes; - Processor structure, functions, and pipelining; - Instruction level parallelism and its static and dynamic exploitation; - Distributed memory hierarchy; - Multiprocessors. 2. Given a set of functional and non-functional requirements, the student can select the most appropriate architecture among a set of different architectures of modern microprocessors. 3. She/he can optimize code for a particular processor using, e.g., code scheduling and loop unrolling. 4. She/he can perform design space exploration and quantify design decisions in terms of performance, energy consumption, cost, flexibility, programmability, predictability for various processor and multiprocessors building blocks and architecture features, e.g., instruction set, predication vs. speculation, cache vs. scratch-pad memory, in-order/out-of-order execution, message passing vs. shared memory, etc. 5. She/he can study recent advances in computer architecture, classify research papers, and report about them.	
Education Method	Lectures, lab work, and reading assignment.	
Literature and Study Materials	J.L. Hennessy and D.A. Patterson, Computer Architecture: A Quantitative Approach, 4th edition, Morgan Kaufman, 2007, ISBN 0123704901. A set of high-impact articles and presentations on computer architecture trends and practices available on Brightspace. IEEE Explore	
Assessment	he lab work and assignments contribute to the final grade as follows: Lab work 40% of the final grade, Reading assignment 60 % of the final grade. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Remarks	Please note that the study guide information may change depending on the developments around the Coronavirus induced situation.	

CESE4090	Reconfigurable Computing Design	5
Responsible Instructor	Dr.ir. J.S.S.M. Wong	
Responsible Instructor	G.K. Kuzmanov	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	System Design with HDLs (ET4272 or CESE4095) or similar ET4170/CESE4035 Computer Arithmetic and ET4171/CESE4040 Processor Design Project	
Course Contents	This course introduces the students to the field of Reconfigurable Computing. The course will provide a balanced insight of both theoretical trends and practical hands-on experience with reconfigurable technology - Field-programmable gate arrays (FPGAs) is the most prominent nowadays. In addition to the basic concepts, the students are taught advanced topics in reconfigurable technologies, organizations, architectures, programming paradigms, design tools and runtime systems. The particular topics to be covered are, but not limited to: fine and coarse grain reconfigurable technologies, partial and runtime reconfiguration, context switching, runtime algorithms for reconfigurable resource management, tools and methods for system-level and RTL synthesis, reconfigurable-specific compiler optimizations. Special attention is paid on application-specific acceleration in the context of reconfigurable technology. For example, the following application domains will be given special attention: image processing, streaming applications, encryption, compression, bioinformatics, supercomputing, etc. The lab attached to this course will focus on high-level synthesis (HLS) that is an increasingly more common way to design for FPGAs.	
Study Goals	- The student understands the basics of reconfigurable technology; - The student is able to design efficiently for reconfigurable technology (in terms of performance and resource utilization); - The student can identify computationally intensive parts of applications that are suitable for acceleration using reconfigurable computing; - The student masters state-of-the-art EDA tools, in particular HLS tools, for reconfigurable computing; - The student will be able to approach the design problems using a top-down methodology (starting from an application, to an architecture, down to a (re)configuration bitstream);	
Education Method	Lectures and laboratory work	
Literature and Study Materials	Handouts and research papers	
Prerequisites	CESE4085 (ET4074)	
Assessment	Open questions written test (50%) + Lab work (50%)	
	disclaimer: information may change depending on the developments around the coronavirus.	

CESE4095	System Design with HDLs	2
Responsible Instructor	Dr.ir. J.S.S.M. Wong	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	Computer Arithmetic and Reconfigurable Computing Design	
Expected prior knowledge	Basic course on logic design.	
Course Contents	As system design often requires the utilization of hardware description languages we concentrate on such a language, i.e., VHDL and their associated simulation and synthesis tools. This course provides students with the background one may require in order to understand, modify, develop and debug VHDL system designs. Covered issues are related to VHDL language constructs as well as to the utilization of simulation and synthesis tools. The addressed topics include among others the following: hardware modeling, simulation, and synthesis; behavioral and component descriptions; signals and entities; delay models; VHDL language constructs; basic I/O; identifiers, data types, and operators.	
Study Goals	The study goals of this course are the following: - the students understand the theory behind hardware description languages - the students are accustomed to the general syntax of VHDL - the students are able to describe simple combinational and sequential logic circuits - the students are familiar with at least one VHDL simulation program, i.e., preferably Modelsim - the students are capable of describing a simple 2-stage processor pipeline design using behavioral VHDL - the students are capable of describing a 32-bit modified Baugh-Wooley multiplier design using structural VHDL - the students are capable to explain their designs and VHDL modeling choices	
Education Method	The theory of hardware description languages and syntax of VHDL are transferred via lectures. In the same lectures, code examples are given of simple combinational and sequential logic circuits. In a lab, hands-on experience with the Modelsim simulator is gained via a tutorial and several simple exercises are given to students to prepare them for the larger two assignments. The larger two assignments require the students (in groups of 2) to give a behavioral description of a 2-stage processor pipeline and a structural description of a 32-bit modified Baugh-Wooley multiplier. Both assignments are assessed via an oral exam in which students have to present their designs and relate them to specific VHDL code sections.	
Literature and Study Materials	Handouts	
Assessment	* Attendance of the lab is obligatory in which students have to follow a tutorial and finish some small exercises. * The final pass/fail course result will be based on an oral exam in which two assignments will be assessed in detail. The assignments can be done by a pair of students.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CS4215	Quantitative Performance Evaluation for Computing Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Today's computing systems become ever complex, due to the rapid development of hardware and software technology. It is challenging to design and run computing systems that guarantee users performance requirements in a resource efficient way. Various quantitative methods are applied to capture such complex system dynamics and predict metrics of interests, from the designing phase of the systems to the runtime performance, e.g., job response times and system anomaly. To optimize the performance of computing systems, a deep understanding on those methods and their applications on the system design are essential. Having practical hand-on experience on designing experiments, deriving models, and validating results with benchmark systems will prepare students to tackle challenges of real world computing systems.	
Study Goals	LO1. Design full/fractional factorial experiments for multi-variate regression analysis, e.g., finding critical parameters for deep learning clusters LO2. Apply queueing theory to analyse and predict run-time performance of applications, e.g., the average response times of on-line ML training service LO3. Apply machine learning models to analyse and predict the system dependability, e.g., root cause analysis for machine failure. LO4. Conduct experiments to profile applications and extract their workload parameters on real systems, e.g., e.g., deep learning clusters LO5. Develop resource management policies and validate them on real computing systems, e.g., deep learning clusters	
Education Method	Lectures: 7 weeks X 2-4h Practical: Derive, validate and evaluate performance models and resource management strategies for a chosen system via homework and group project. Multiple types of computing and network systems can be chosen from. Deliverables include a report and group presentation.	
Books	1) Performance Modeling and Design of Computer Systems: Queuing Theory in Action by Mor. Harchol-Balter 2) Design and Analysis of Experiments by Douglas C. Montgomery, 3) The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer Series in Statistics.	
Assessment	Homework (40%): 2 individual homework Group project (60%): group project report and presentation	
disclaimer: information may change depending on the developments around the coronavirus.		

EE4610	Digital IC Design	3
Responsible Instructor	Dr. S. Du	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear circuit theory. Digital circuits basics. Helpful: prior / BSc level knowledge of CMOS technology and design	
Course Contents	This course will present a broad yet thorough overview of the subject of digital VLSI design, spanning both circuit and system abstractions. This complete picture is the only way to make the right tradeoffs, find the most suitable optimizations and the best implementation strategies for very large scale integrated circuits in standard deep-submicron CMOS technologies. After an introduction to technology, devices and interconnect, combinational logic gates and sequential elements are studied. This is followed by system level perspectives of interconnect issues, timing issues and the design of macro blocks. At each level, the opportunities and limitations of the physical implementation are considered for finding better solutions and trade-offs. This includes the consequences of the analog behavior of digital systems with respect to e.g. cross-talk noise and signal waveforms, that generally tend to become more influential with each new technology generation.	
Study Goals	Be able to analyze, design, and verify a VLSI component with full comprehension of how its performance, power dissipation, size and reliability relates to the physical implementation. Be able to use this knowledge to make suitable tradeoffs and implementation choices. Be able to manage the complexity of scale of a VLSI system as well as the complexity of behavior of its individual components.	
Education Method	Lectures and design assignment	
Computer Use	Practice assignment (Spectre and Cadence)	
Books	"Digital Integrated Circuits: A Design Perspective", 2nd edition, J.M. Rabaey et al., Prentice Hall / Pearson, 2003.	
Assessment	Two assignments, Written examination, open book Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	Dr.ir. N.P. van der Meijs	

EE4615	Digital IC Design II	3
Responsible Instructor	S.M. Alavi	
Responsible Instructor	M. Babaie	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course will consolidate the knowledge acquired in EE4610 by practicing the design of an advanced complex digital VLSI design, where high-level specifications are translated to transistor sizes, going through several abstraction levels. At each level, the opportunities and limitations of the physical implementation will be considered for finding better solutions and tradeoffs, by means optimization and the search of new topologies, while keeping complexity to a minimum to achieve the desired functionality. Parasitics and the other effects deriving from the analog behavior of digital circuits will be taken into account using four-corner analysis, possibly Monte-carlo analysis, as well as simplifications and modeling at each abstraction level and other semi-automatic tools.	
Study Goals	The main goal of the course is to give students the tools to successfully design complex, scalable circuits and systems. The students will also learn how to work in and coordinate teams of designers around a complex project from high-level specifications to the actual transistor-level implementation and post-layout verification.	
Education Method	The course will be entirely based in the lab where the target design will be implemented by students in groups of 2-3. Only based on students/groups wish, the final design will be laid out in a commercial CMOS process and possibly fabricated and tested.	
Assessment	The course will be graded based on the target design. The following criteria will be taken into account: 1. Functionality achieved; 2. High-level specifications met; 3. Four-corner analysis (Monte-Carlo analysis is optional)completed; Bonus points will be given to those students that will exceed the specifications and create a robust, reproducible design. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

EE4690	Hardware Architectures for Artificial Intelligence	5
Responsible Instructor	Dr.ing. R.K. Bishnoi	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Basics of Digital design	
Course Contents	Recent advancements in the field of Artificial Intelligence (AI) have demonstrated its great potential in solving exceptionally complex problems, that can even surpass humans. To support such AI algorithms, the demand of developing new and efficient hardware is growing with rapid pace. In particular, for a variety of edge/IoT devices, hardware platform locally deploys these algorithms that decreases the exchange of information with the cloud which significantly reduces the networking costs and boost the overall performance of the devices. Hardware architectures play a critical role in executing AI algorithms. This course gives an overview of AI from hardware perspective and discusses close to the state-of-the-art architectures and training mechanisms. In this course, you will learn the concept of advanced forms of AI (i.e. Deep Neural Networks) and how they can solve the complex problems. You will also learn, what are the challenges associated with the underlying hardware and how new architectures can address those. Moreover, the course will teach you the concept of brain-inspired computing using in-memory processing hardware platform. Technically, this course covers following topics: 1) Machine inference models (i.e. Support Vector Machine, Decision tree, Regression analysis, clustering, etc.), machine learning algorithms (i.e., Supervised, Unsupervised, Semi-supervised, Reinforcement learning models) and their associated challenges. 2) Deep Neural Network (DNN), Classification of DNNs, feed-forward, back propagation and learning mechanism (such as Gradient decent and Stochastic learning). 3) Spiking Neural Network, Spike Time Dependent Plasticity training algorithms and some existing hardware architectures (such as Intels Loihi, SpinNaker, IBMs TrueNorth, Neurogrid, etc.). 4) Emerging technologies for neuromorphic computing (using memristor devices such as RRAM, MRAM & PCRAM) and non-conventional computing paradigm (such as computing-in-memory, hybrid-design approach, crossbar arrays).	
Study Goals	The main aims of the course are: 1. Describe the concept of AI, inference and training models as well as their associated challenges from hardware perspective. 2. Understand the hardware AI requirements for the data-centers as well as edge devices. 3. Discuss the neuromorphic computing and its advantages with respect to conventional computing paradigms. 4. State details about Artificial Neural Network and their training algorithms. 5. Develop understanding about Spiking Neural Network and their training mechanisms and also, comparison with Artificial Neural Network and their evaluations based on design metrics such as accuracy, power, speed, size, etc. 6. Describe state-of-the-art neuromorphic hardware architectures and discuss their design implementation details. 7. Study the advanced topics of neural networks based-on computation-in-memory concept using emerging non-volatile memories. 8. Become an AI engineer with good understanding about hardware implementation.	
Education Method	Lectures, assignments and lab-work	
Assessment	Oral and lab	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Elective	Yes	
Tags	Artificial intelligence	

EE4695	Hardware Dependability	5
Responsible Instructor	M. Taouil	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	0,0,4,0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A course on VHDL/Verilog design such as one of the following: - Digital Systems B (EE1D21) - System Design with HDLs (ET4272)	
Course Contents	Integrated circuits (ICs) are integral part of any electronics system we see around. Developing, designing and deploying of such chips is by far more than just getting your SPICE simulation showing a correct behavior. Ensuring high quality, reliable and secure manufactured chips is a critical and crucial step in the whole IC process. A chip targeting children toys could cost 0.5 while a similar chip (in terms of functionality) could cost 100X to 1000X more when used for critical application such as military. Hardware dependability (e.g., quality, reliability, security) is a key differentiator. This course will introduce you to hardware dependability concepts and aspects. You will learn that ensuring such dependability needs the design of appropriate countermeasures, both during the design phase (i.e., part of the design) and in field/ during the lifetime of the application (i.e., continue monitoring and acting when needed). The course will teach you how to guarantee that a chip that may consists of billions of transistors and connections can be tested in less than one second, why the warranty you get for your phone is just two years rather than e.g., 5 years, how hackers can retrieve secret information from your device or smart phone and how to prevent it from happening, etc. Technically, the topics that will be covered are: a) Test/Quality: manufacturing defects, fault modeling, test development, design -for-testability, memory test, build-in-self test, etc; b) Reliability: reliability failure mechanism, modeling, design-for-reliability, testing, etc.; c) Security: hardware attacks and countermeasures, IP attacks, functionality Attacks, Data Attack, design-for-Security, etc.	
Study Goals	The main aims of the course are: 1. Describe the importance of Hardware Dependability (i.e., testability, reliability, security, etc.) and understand its relevance to electronic design and its overall impact . 2. Point out the strong correlation between VLSI Design, Test, Reliability and security 3. State the trends and challenges in test, reliability, security domain 4. Describe transistor and interconnects defect mechanisms and the way they behave at the electrical/functional level and how they are tested using fault models and test algorithms 5. Examine different test methodologies for logic and sequential circuits, their advantages, disadvantages, cost, limitations, etc and how they impact the security 6. Analyze and propose different test algorithms and Design-for-Testability DFT methodologies, their advantages, disadvantages, cost and limitations and how they impact the security 7. Implement block ciphers in hardware and evaluate their weaknesses 8. Analyse hardware attacks and evaluate the countermeasures that can be used against them 9. Develop fault injection and side channel attacks and design countermeasures for them 10. Become a better VLSI designer, a better test engineer/ product engineer, and a better hardware security engineer.	
Education Method	Lectures, homework assignments and lab	
Literature and Study Materials	- Lecture Notes - Lab manual - Book: Essentials of Electronic Testing (by M.L. Bushnell and V.D. Agrawal, ISBN 0-7923-799-1-8) - Syllabus containing material from online resources and material from the following books: - Introduction to Hardware Security and Trust - Trustworthy Reconfigurable Systems - Trusted Computing for Embedded Systems - Security in Embedded Devices	
Assessment	Assignments + lab + "group" oral examination disclaimer: information may change depending on the developments around the coronavirus.	

EE4700	Modelling, Algorithms and Data Structures	5
Responsible Instructor	Dr. S. Feld	
Instructor	Prof.dr.ir. S. Hamdioui	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	- Basics of digital technology - Boolean logic and reasoning - General concepts of programming languages	
Course Contents	To solve a given electrical engineering (EE) problem, we usually start with developing an appropriate model of such a problem. Once the problem is modeled using an appropriate data structure, one can focus on solving the modeled problem instead of the EE problem itself. Choosing the right data structure is therefore important for generating efficient and optimal solutions. Solutions for such problems consist usually of algorithms. Analyzing the algorithms in terms of efficiency, time complexity, required memory, etc. is therefore an important aspect when dealing with Electrical Engineering problems. This course deals with these three important aspects of solving electrical engineering problems, namely modeling, algorithms and data structures. It will cover the following topics: sets and logic; proofs; functions, sequences and relations; algorithms and complexity; introduction to number theory; counting methods; recurrence relations; graph theory; trees; network models.	
Study Goals	By the end of the course, the students are able to: - LO1: Create models for specific problems - LO2: Proofs a) Classify different proof methods b) Use them to prove theorems - LO3: Algorithms a) Develop algorithmic solutions for given problems b) Analyse those algorithms - LO4: Apply counting to discrete problems - LO5: Employ graph theory to solve some problems	
Education Method	Lectures	
Literature and Study Materials	- R. Johnsonbaugh, Discrete Mathematics, Pearson Prentice Hall, 7th Edition. - Lecture slides and other material, such as scientific papers, will be distributed.	
Assessment	Written exam.	
Disclaimer: information may change depending on the developments around the coronavirus.		

ET4351	Digital VLSI Systems on Chip	4
Responsible Instructor	Dr.ing. C.P. Frenkel	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basics of digital design. Previous exposure to hardware description languages (Verilog or VHDL) is strongly recommended.	
Course Contents	Today's large-scale digital integrated circuits and systems-on-a-chip, such as modern general processing units (GPUs), can embed no less than 50 billion transistors. Two things were instrumental in enabling this massive scalability: electronic design automation (EDA) tools and intellectual property (IP) blocks allowing for hierarchical design. In this course, you will learn how to use such EDA tools and IP blocks to go from the register-transfer level (RTL) description of your digital design to a final layout that is clean for tapeout and meets your specifications. To do so, you will also learn about power/performance/area tradeoff optimization, hardware-algorithm co-design, and memory hierarchy. These tools and concepts will be practiced and applied in a project, in which you will explore and apply your own design choices to meet a well-defined design target in a digital system-on-chip.	
Study Goals	With this course, you will get familiar with digital system-on-chip integrated circuit design using EDA tools. The main goals are to: 1) understand and apply automated synthesis and place-and-route tools, 2) be able to explain and optimize the power/performance/area tradeoff to meet the specifications of a digital design, 3) carry out a proper IP selection to meet these design targets, 4) understand how algorithmic requirements link to hardware constraints, 5) verify a digital design and demonstrate that (i) simulation results can be trusted, and (ii) that the design is clean for tapeout.	
Education Method	Lectures, labs	
Literature and Study Materials	* Course material (slides will be provided) * (Recommended) Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, by Erik Brunvand, Addison-Wesley, ISBN-10: 0321547993	
Assessment	The final course grade consists of two parts: * Final design report (70%) * Individual oral exam (30%) The results of the two parts must have been obtained within the same academic year, and students must have a 5.8 or higher for both the final design report and the individual oral exam to pass the course. In case either of the two parts is failed, a resit must be taken in the form of an individual oral exam. Depending on the part(s) that were failed, the oral exam will re-assess the course concepts and/or project-related aspects. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	C. Gao	

ET4362	High Speed Digital Design for Embedded Systems	5
Responsible Instructor	Prof.dr.ir. G. Gaydadjiev	
Assistant	Dr.ir. C. Strydis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The course covers many important topics computer engineers and embedded system designers should be aware of. First, the physical effects of high-frequency signal propagation and transport are discussed. Next, the state of the art for high bandwidth/low latency digital communication and memory systems is presented. Special attention is given to parallel memory systems. Thereafter, low level circuit design and analysis principles relevant to high-speed digital design are explained. Clock distribution, metastability and skew-tolerant design are emphasized. Advanced techniques for high-speed combinational and sequential logic structures are considered in detail. More precisely, approaches like domino logic, pulsed latches and dual-rail logic will be discussed. The next part discusses advanced design techniques such as globally asynchronous locally synchronous (GALS) systems, systolic arrays, streaming dataflow, wave-pipelined circuits and fully asynchronous design. The last part concentrates on understanding the true theoretical bounds in, e.g., the energy consumption, of computers. The necessary theory and some practically realizable adiabatic logic design styles will be introduced. When time allows, brief introduction to the Gödel's incompleteness theorem and its impact on computers will be presented.	
Study Goals	The purpose of this course is to familiarize students with the latest developments in the field of high-speed digital design. The emphasis is on novel, non-traditional design concepts, techniques and methodologies.	
Education Method	Lectures and individual assignments.	
Literature and Study Materials	Presentations and selected research papers.	
Assessment	Written report on a selected research topic and oral exam on appointment.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IN4350	Embedded Computer Architectures 2	5
Responsible Instructor	Dr.ir. A.J. van Genderen	
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	N.A.	
Course Contents	Tasks to be executed by digital hardware become more and more complex. For that reason, it is desirable to have a structured design process to come from task (algorithm) to hardware. Besides that, powerful concepts to express the algorithms in a formal way are required. This course aims at providing in-depth knowledge of the relation between an algorithm and the possible architectures that execute the algorithm. We will discuss methods to transform a sequentially expressed algorithm via single assignment code and a dependency graph into a signal flow graph. From there on various hardware implementations will be developed in the form of array processors or systolic arrays. In case of an irregular dependency graph, the hardware will take the form of a simple von Neumann architecture with a dedicated instruction set. Topics included are: - Linear (static) scheduling - Pipelining - Re-timing and design transformations. A second aim of the course is to specify the resulting hardware alternatives in a formal way, by means of mathematical equations. Using the evaluation mechanisms of a functional programming language, these equations can be directly evaluated leading to a simulation of the hardware. Besides, these specifications open the possibility to prove that the functionality of the various hardware alternatives is the same. Theory concerning the structured design process and the specification of hardware in a formal way will be discussed during lectures. The students gain hands-on experience in a project assignment where an algorithm is described in a formal way and synthesized toward hardware. See also https://osiris.utwente.nl/student/OnderwijsCatalogusSelect.do?selectie=cursus&cursus=192130250&collegejaar=2022&taal=en	
Study Goals	The student after completing the course will be able to: - derive recurrent relations from mathematical expressions; - design a data dependency graph based on the recurrent relations; - transform the data dependency graph into a signal flow graph using projection and scheduling; - describe mathematical expressions that specify data dependencies and operations in a graph by means of a functional programming (FP) language; - synthesize hardware circuits using the FP descriptions; - describe the results of the project assignment in a report.	
Education Method	This course is provided remotely by the University of Twente through tele-lectures and homework assignments.	
Literature and Study Materials	Will be made available via the CANVAS internet platform in Twente University	
Assessment	Written examination	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Embedded Systems

Specialisation courses Networking (choose at least 15 EC) 2022

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet has become of critical importance to society. However, the large size of networks and abundance of protocols have made network management very complex. The novel concept of network programmability addresses this complexity and has resulted in a paradigm shift in how networks are (or can be) operated.	
Study Goals	The high-performance data networking course is an advanced networking course that will introduce you to the concept of network programmability and which treats fundamental networking concepts like Quality of Service and network resilience.	
	The learning objectives of this course are twofold: (1) The student should gain knowledge of the treated networking technologies. (2) The student should be able to apply and work with the programmable network technologies in a network emulator (Mininet).	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final assessment will be based on an exam that covers both the theory from the slides as well as the content from the reader.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4050	Measuring and Simulating the Internet	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	(Advanced) Networking course (e.g., CS4055/CESE4045) and Programming skills.	
Course Contents	The Internet is a complex network without a fixed structure. Hence, measuring the Internet is crucial to acquire knowledge about the Internet infrastructure (topology), traffic, and performance (e.g., loss, delay, bandwidth, etc.). This course will discuss the design requirements and challenges in measuring and simulating the Internet, and the existing measurement methodologies (how/where/when to measure). Knowledge of how to conduct and evaluate Internet measurements enables the design and enhancement of a large set of applications, including: capacity planning and traffic engineering, network management and trouble-shooting, detecting network abuse and intrusions, etc.	
Study Goals	The goal of this course is to introduce the students to basic Internet measurement tools, as well as the state-of-the-art in Internet measurements research. The students will learn several Internet measurement techniques (e.g., active vs. passive measurements), and different software tools. Through a measurement assignment, the students will learn how to define/formulate a research problem, choose a specific approach, and complete a measurements-related research project.	
Education Method	Lectures + weekly instructions (discussion of papers and progress) + independent project work.	
Literature and Study Materials	Papers	
Assessment	Groups of students will be assigned a project that requires the students to put the theory on measuring and simulating the Internet into practice. The students have approximately 1 month to complete their assignment. The final assessment is based on the presentation (via report and/or demonstration) of the project assignment results and on the individual contribution and level of participation. Students within a group may thus receive different grades. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Maximum number of participants	Because this is a project-based course, we can only admit a limited number of students (typically around 30, but the actual number depends on the number of TAs involved). If more students enrol, we will give preference to those who have successfully completed CS4055/CESE4045.	

CESE4055	Ad-hoc Networks	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ----- Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. ----- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) - o Definition of ad-hoc networks - o Comparison with infrastructure based systems - o Typical applications - o Advantages and challenges - o Radio technologies for ad-hoc networks - o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks - o Propagation models - o Topology models based on graph theory - o Degree and hopcount - o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) - o Introduction to MAC protocols - o Issues and design goals - o Classification - o Directional, multi-channel MAC protocols - o Energy efficiency in MAC protocols - o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) - o Flooding - o Node discovery, neighbour discovery - o Route establishment - o Topology maintenance, localisation - o Proactive, reactive and hybrid routing - o Typical protocols - o Energy efficiency in routing - o Broadcast and multicast - o Effects of mobility on connectivity and capacity - o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks - o Wireless sensor networks (Chapter 12 of the textbook and papers) - o Cooperation (Reference papers) - o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) - o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consisting of slides presented at the lectures (Slides are only teaching aid and they are not substitute for	

textbooks, research papers, etc).
3. Some recent journal papers

4. Optional Reference Books

- 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms)
- 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar , C Puttamadappa , and T. G Basavaraju, Auerbach Publications, 2008. This book is available online in the library.
- 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is available online in the library.
- 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer.
5. OPNET/ns-2 web pages, tutorials and video lectures

Books

Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.

Assessment

1. There will be totally three written/oral tests/examinations for this course (including the final exam).
2. The students will carry out a project in a group and submit a short report.
3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis.

Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class.

Breakup:

2 Tests + Final Exam = 55%

Project 40%

Self assessment + Reflection 3%

Activities on Feedback Fruits or any Online platform 2%

Resit: written/oral exam [+ carry over from projects].

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Changes due to COVID19 in 2021 is below;
in case there is a similar situation, we use the following methods.

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There are three chapters

Chapter 1: Network modelling

Chapter 2: MAC protocol

Chapter 3: Routing

+

Lab: Simple experiments using laptops (individual) + simulations (group-wise)

Marks Breakup

Homework/Assignments/Group works

Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks

Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks

Part B1: Individual experiment + report

Part B2: Group-wise simulations + report + demo

Part B1 + Part B2 - 50 Marks

Part C: Self-assessment + peer activities -- 5 marks

Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit.

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(More information will be given in the first class)

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)

Permitted Materials during Tests

Different conditions for different test/exams.

Conditions will be informed 1 week before the exams/test.

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	DISCLAIMER: this study guide information may change depending on the developments around the corona virus. The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Books	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/ ; Wireshark learn page: https://www.wireshark.org/#learnWS ; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/ ; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course.	
	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

CESE4065	Advanced Practical I.o.T. and Seminar	5
Responsible Instructor	R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.	
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break	
Summary	his course is an involved hands on course for self motivated students. Students work in a group of two (max 3) usually. Students are expected to have sufficient programming and hardware development skills. The course is project oriented thus students are expected to deliver a working model with demonstrations. Preference will be given to Q5 students. Q1 students are discouraged unless otherwise explicitly allowed. ----- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. ----- This course is done in a group of two (max 3). Instructor/TA will help connecting a potential groupmate in the first class using Brightspace/Feedback Fruits. ----- Activities: How to in Groups? ----- Part 1: Seminar We guide the groups to select a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID19 related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their own responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/raspberry pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki based networks or Open testbeds projects. In these cases, online group activities are possible. NOTE: In some extreme cases, Instructor/TA may allow one person project after assessing the situation and after discussing with the student. ----- Supply of Hardware: ----- 1. Instructor/TA is able to provide take home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components.	
Course Contents	Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.	
Study Goals	To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.	
Education Method	Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).	
Literature and Study Materials	This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.	
Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc.	

	Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project. Part 1: 30% of the whole marks; Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment. Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference. Report will not have individual component, however, Q/A may have individual component. Please NOTE: There would be no resit since this is a hands-on course being executed in a team. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)
Tags	Circuits Group work Programming Project Software
Judgement	30 Marks for Seminar 60 Marks for project, demonstration, Q/A 10 Marks for making a paper that could be submitted as a manuscript for review to a conference/journal. Bonus marks are offered for excellence in various components during the course. That would add to the total marks not exceeding a max of 100.
maximum aantal deelnemers	Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (Fundamentals of quantum information)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution. Caution: 1. This class requires you to take "Fundamentals of Quantum Information" in Quarter 1 2. The focus of this class is presently on quantum communication, and we will only briefly look at quantum cryptography. As such, this class is not held in flipped classroom style in conjunction with edX QuCryptoX as in previous years.	
Study Goals	The student will acquire: - A good understanding of the fundamental concepts of quantum communication - Insight into the differences between classical and quantum communication and cryptography - Skill set required to follow the remainder of the quantum curriculum (Q301 Quantum hardware and Q401 Quantum electronics)	
Education Method	Lectures and tutorials. If remote classes continue this fall, then recorded lectures and live discussion session	
Literature and Study Materials	Primary: - Slides - Review Articles Auxilliary: - Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	Homework (70%) and Final Project (30%)	
Tags	disclaimer: information may change depending on the developments around the coronavirus. Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Linear Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4215	Quantitative Performance Evaluation for Computing Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Todays computing systems become ever complex, due to the rapid development of hardware and software technology. It is challenging to design and run computing systems that guarantee users performance requirements in a resource efficient way. Various quantitative methods are applied to capture such complex system dynamics and predict metrics of interests, from the designing phase of the systems to the runtime performance, e.g., job response times and system anomaly. To optimize the performance of computing systems, a deep understanding on those methods and their applications on the system design are essential. Having practical hand-on experience on designing experiments, deriving models, and validating results with benchmark systems will prepare students to tackle challenges of real world computing systems.	
Study Goals	LO1. Design full/fractional factorial experiments for multi-variate regression analysis, e.g., finding critical parameters for deep learning clusters LO2. Apply queueing theory to analyse and predict run-time performance of applications, e.g., the average response times of on-line ML training service LO3. Apply machine learning models to analyse and predict the system dependability, e.g,root cause analysis for machine failure. LO4. Conduct experiments to profile applications and extract their workload parameters on real systems, e.g., e.g., deep learning clusters LO5. Develop resource management policies and validate them on real computing systems, e.g., deep learning clusters	
Education Method	Lectures: 7 weeks X 2-4h Practical: Derive, validate and evaluate performance models and resource management strategies for a chosen system via homework and group project. Multiple types of computing and network systems can be chosen from. Deliverables include a report and group presentation.	
Books	1) Performance Modeling and Design of Computer Systems: Queuing Theory in Action by Mor. Harchol-Balter 2) Design and Analysis of Experiments by Douglas C. Montgomery, 3) The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer Series in Statistics.	
Assessment	Homework (40%): 2 individual homework Group project (60%): group project report and presentation	
disclaimer: information mav change depending on the developments around the coronavirus.		

CS4430	Network Security	5
Responsible Instructor	Dr.ing. A. Zarras	
Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best practices in computer and network security. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course's primary focus will be on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design, review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics, as well as conducting measurements on the effectiveness of attack and defense schemes.	
Study Goals	See course contents.	
Education Method	Lectures, Labs, and Project.	
Assessment	Assignments and Project.	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

EE4396	Mobile Networks	5
Responsible Instructor	Dr. R. Litjens	
Contact Hours / Week x/x/x/x	0/0/0/5	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Not formally required, but recommended to have the basic knowledge taught in 'Fundamentals of Wireless Communications' (ET4358). If you didn't take this course, key aspects will be repeated during the class. In case you are lacking a comfortable background, talk to me in the first week of class and I'll help you catch up.	
Course Contents	The course addresses the technology and management of contemporary and future mobile cellular communication networks. With a focus on radio access, as it dominates both the performance and cost aspects of mobile networking, the course addresses 5G, 4G (LTE/LTE-A), 3G (UMTS/HSPA) and 2G (GSM/GPRS/EDGE) technologies, all of which are operational network technologies. An outlook to 6G will also be given. Fundamental aspects of cellular networking are explained and discussed, including e.g. the intrinsic tradeoffs between coverage, capacity and quality and key aspects that influence performance (traffic, propagation, technology, deployment, spectrum. Cellular network planning, dimensioning and performance analysis are treated, as well as radio resource management mechanisms (adaptive beamforming, channel-aware single/multi-user scheduling, handover control, link adaptation, admission, congestion control,) and the current trend towards (potentially AI/ML-based) self-management. Cutting-edge radio features including single/multi-user massive MIMO beamforming, which is at the heart of 5G, are also covered. Considering the on-going technological developments in this exciting domain, the course material is updated every year. The principal lecturer has a 25+-year track record of research and consultancy in the field. The course is enriched with notes about actual deployments, historical anecdotes, popular misconceptions, fascinating factoids, illustrative demonstrations and (potentially) guest appearances.	
Study Goals	The objective of the course is to provide the student with a thorough understanding of contemporary mobile cellular networking technologies, their fundamental properties and tradeoffs, as well as the key challenges and approaches related to network dimensioning, planning and (self-)optimization towards efficient provisioning of coverage, capacity and quality.	
Education Method	Lectures, demonstrations, in-class discussions, assignment(s).	
Literature and Study Materials	Primarily: the lecture slides! For the enthusiastic student, supplementary background materials are recommended in the form of e.g. (white/scientific) papers and book chapters.	
Books	A number of books on wireless/mobile communications in general or about 2/3/4/5G technologies specifically are recommended as background material; specifics will be given in class / in the lecture slides. As said, this is merely recommended for the most eager students. In principle, the lecture slides cover all relevant content.	
Assessment	If COVID regulations comfortably allow: written, closed book exam; otherwise: oral, closed-book exam. And one (may be two) assignments.	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Prof.dr.ir. P.F.A. Van Mieghem	

EE4630	Telecommunication Network Architectures	3
Responsible Instructor	Ir. R.A.C.J. Noldus	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course presents the fundamentals of cellular mobile networks, including network architecture and services, with a focus on end-to-end and core network aspects. The course forms a companion course for Cellular Networks Radio Access Networks (ET4396). Students will be introduced to the concepts circuit switched and packet switched, the two fundamental communication methodologies that form the backbone of the mobile network. The course aims to educate students on the evolution of the network from 2G to 5G, whereby for each network generation ('G'), the new architectural principles are studied. For example, we discuss the GSM core network (2G), GPRS (2.5G), UMTS (3G), Evolved Packet Core (EPS; 4G), the IP Multimedia Subsystem (IMS), Multimedia telephony (MMTel), Intelligent Networks (IN), Circuit switched fallback (CSFB), Voice over LTE (VoLTE), Voice call continuity (VCC), Voice over Wifi and 5G network architecture. For the 5G network architecture, the concept of 'network slicing' will be presented, which is used for flexible resource allocation, for Internet of Things (IoT) applications and for 'verticals' such as Automotive, Smart Energy, Media & Entertainment, Vehicle communication and other forms of machine-type communication. The course also deals with the concept of virtualization and cloud computing. Networks are increasingly built as a collection of virtual network functions (VNF), deployed on industry standard computing platforms, comprising standard hardware and standard OS. Keywords: - Network architecture and network protocols - Core network, Radio access network (RAN) - Circuit switched, Packet switched - Location/mobility management/roaming - Security (authentication, ciphering,) - QoS/QoE/bearer architecture, QCI definitions, service management, policy control - IMS/VoLTE/Messaging/Voice-over-Wifi/Enriched calling - Virtualization, Cloud computing, Software Defined Networks (SDN) - Network slicing - Multi access Edge Computing (MEC) - Intelligent Networks - Verticals/applications, including e.g. Automotive, Smart Energy, IoT, Machine-to-machine communication - Web Real-time Communication (WebRTC)	
Study Goals	After this course, students are able to explain the meaning, working and rationale behind key cellular network concepts (such as those given by list of keywords above). Furthermore, students will be able to understand the challenges and associated trade-offs in the deployment and management of cellular networks. Students will also be able to analyse network architecture and evaluate network architecture on its merits or on its drawbacks.	
Education Method	This course consists of 8 classes with each class consisting of 2 hours. After the 4th lecture, a mid-term rehearsal takes place; and after the 8th lecture, an end-term rehearsal take place. These rehearsals help the students in preparing for the exam. The students are expected to study the lecture material after every lecture and to ask questions the next lecture, on any topic that requires further explanation. Homework will be provided after each lecture. Students may at any time interact with the lecturer via mail, to ask clarification on any topic of the course.	
Literature and Study Materials	Slides Articles and book chapters	
Assessment	Examination, closed book.	
Co-Instructor	Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4) Prof.dr.ir. P.F.A. Van Mieghem	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework	
Assessment	written examination	
Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)		

ET4034	Telecom Business Architectures and Models	4
Responsible Instructor	Dr. E.F.M. van Boven	
Responsible for assignments	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	none	
Summary	Edgar van Boven, (KPN, TUD EEMCS)-- Past, present & future of communications -- Assignment 1 Eric Smeitink (KPN, TUD EEMCS)--Strategy, Spectrum & 5G Ton van der Knaap, (Stedin)-- What, why and how of business architecture; experiences from a grid operator and a telecom operator compared Samuel Pronk, (Krauthammer)-- Real Options-- Assignment 2 Ramin Hekmat, (KPN)-- Security and privacy in contemporary communication networks-- Assignment 3 Wouter de Vries (KPN)-- Artificial Intelligent Processes for People Tim Daeleman, (Prodapt Consulting)and Erik Lemmens (KPN)-- Digital Telecom Platforms Jose Almodovar, (TNO)--Standardization of Mobile Communications Frank Mertz (KPN)--5G Networks Developments and Opportunities Remco Helwerda (KPN) -- Digitalization Marco van der Pal, (Cap Gemini)--The IoT landscape architecture Ruud van de Bovenkamp (KPN)-- Fiber-to-the-Home Access Networks -- Assignment 4 Mark Dirksen (KPN)-- Business Continuity Management Pieter Veenstra, (Titanium)-- Service Architecture and 5G Roaming Security Edgar van Boven (KPN, TUD EEMCS)-- Value Case pitch practicing -- Assignment 5	
Course Contents	The essence of the course The ET4034 course Telecom, Architectures and Business Models was designed in 2001 and scheduled since 2002. The aim of the course is giving students from any faculty of the Delft University of Technology the opportunity to learn about a large variety of aspects of the telecom domain, not only its technologies, but also the diverse and challenging daily practice of organisations active in the communications sector. The course is provided by a team of 15 lecturers interactively sharing their knowledge about communications infrastructure in the context of overarching societal trends, service & network architecture, long term developments and new technology evolving from standardization initiatives. As a consequence of continuous innovation and societal developments, the course content is being updated each year including a well-considered choice of central themes that enhance and connect the course lectures. In 2022, the themes Internet of Things, Artificial Intelligence and 5G were highlighted, to be updated in 2023.	
Course Contents Continuation	EE5010 Internships or ET4399 projects in telecom related industries.	
Study Goals	Overview and understanding the Communications Sector, Services, Telecom infrastructures and Value Cases serving society	
Education Method	Online lectures, one powerpoint pitch presentation given by student teams and four off-line written assignments in teams of 2 or 3 students	
Literature and Study Materials	See information on Brightspace.tudelft.nl	
Reader	Course lectures, background information and video recordings available on ET4034 Brightspace	
Prerequisites	Finalised BSc as ET4034 is meant for Master students. PhD students are also welcome and can obtain credits when successfully passing the course assignments via the Graduate School.	
Assessment	The examination of the course Together forming the basis for the examination, the ET4034 course contains five assignments of which four written documents and one opportunity to strengthen ones pitching skills presenting a self-developed Value Case to a live audience. Students are invited to work in couples or trio's on the five assignments. The four written assignments each have a weight factor 1 and the Value Case pitch presentation has a weight factor 0,25 in the end mark of the ET4034 course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Exam Hours	Not applicable, the course assignments are four documents written offline together with a course partner. The Value Case powerpoint presentation is a pitch that takes circa 7 minutes.	
Maximum number of participants	No limitation	
Self test	Not applicable	
maximum aantal deelnemers	No limitation	

ET4358	Fundamentals of Wireless Communications	5
Responsible Instructor	Dr.ir. G.J.M. Janssen	
Instructor	Dr. R. Litjens	
Instructor	Dr.ir. J.H. Weber	
Instructor	P. Pawelczak	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Basics on Telecommunication Techniques and Digital Communication Networking.	
Course Contents	The Fundamentals of Wireless Communications course provides a selection of fundamental topics of modern wireless communication systems. The course is prerequisite for the courses on Wireless IoT and Local Area Networks (ET4394) and Mobile Networks (ET4396). The following aspects will be covered in this course. 1. Radio propagation: - review of radio propagation, link-budget model, - path-loss and shadow fading, - multipath propagation: * time domain: rms-delay spread, signal dispersion, inter-symbol interference, * frequency domain: frequency selectivity, coherence bandwidth, - stochastic propagation models. 2. Modern modulation techniques: - review of basic modulation techniques, - Orthogonal Frequency Division Multiplexing (OFDM), - Direct Sequence Spread Spectrum (DS-SS), - Frequency Hopping. 3. Diversity: - diversity domains: frequency/time/spatial/polarization diversity - transmit and receive diversity, - diversity techniques: selection diversity, maximal ratio combining, equal gain combining, - MIMO (multiple input multiple output), fundamentals, capacity, space-time coding. 4. Channel Coding: - characterization of error types, and basics of error control coding, - code design: trade-off between efficiency, reliability, complexity/delay, - block & convolutional codes, - puncturing, interleaving, - Turbo coding. 5. Multiple Access: - multiplexing: FDMA, TDMA, CDMA, (O)FDMA, ALOHA, - delay modeling in multiplexing: Little's theorem, review of continuous time Markov chains, M/M/1 system. 6. Cellular System Design: - key cellular concepts, incl. frequency reuse, sectorisation and interference, - coverage, capacity, quality trade-off, - basic coverage planning using a link budget analysis, - basic capacity planning using Erlang loss/processor sharing models.	
Study Goals	The student has gained understanding and insight in the basic concepts and principles applied in wireless communication systems, as described under Course Contents. He/she is able to explain the relations between the different concepts and trade-offs that can be made and can apply this knowledge by solving related problems by means of calculations.	
Education Method	Lectures, instruction lectures and homework exercises.	
Literature and Study Materials	Slides, notes, and papers, to be provided by the instructors. The following books are recommended as background material: - "Wireless Communications", Andrea Goldsmith, Cambridge University Press, 2005, ISBN 9780521837163, - http://web.mit.edu/dimitrib/www/datanets.html The following book is recommended as refresher material: - "Digital and Analog Communication Systems", L.W. Couch II, Prentice Hall, 2013, ISBN 978-0-13-291538-0.	
Assessment	Written exam (closed book). Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Permitted Materials during Tests	During the written exam it is allowed to use the tables booklet "Tables & Graphs for use at the exam Wireless Communications ET4358", as provided on Brightspace, 1x A4 of your own original hand-written notes/equations (2-sided) (not type-written, no copy of notes and no worked out exercises or examples) and a non-programmable electronic calculator.	
Special Information	This study guide information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Remarks	This course belongs to the set of elective track core courses of the MScEE track Wireless Communication & Sensing and it is an elective course for other MSc programs. Actual course information is available on Brightspace.	

IN4254	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course TI 1206 in our first year Bachelor curriculum "Object Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course TI 2216M in our second year Bachelor curriculum "Probability and Statistics".	
Course Contents	We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming. The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved on research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve the particular problems in their areas. After taking this course students will be able to: (1) Explain the current applications, methods and research trends in the area of smartphone sensing. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non-trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	Written reports + project presentation + oral exam Overall, the final grade is determined by: 1) Two intermediate reports (5% of grade each, 2 pages each) 2) Final report (10 % of grade, 5 pages) 3) Final project demonstration (80% of grade) The first two reports are due on the third and fifth week; and the final report, project and exam are due on the ninth week. There is no resit for this course. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with ES students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot. IMPORTANT: The study guide information may change depending on the developments around the coronavirus.	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Literature and Study Materials	We follow the book Performance Analysis of Complex Networks and Systems, by P. Van Mieghem, Cambridge University Press (2014). See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	Written and closed book. A formulary is provided that can be consulted at the examination. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

SC42100	Networked and Distributed Control Systems	3
Responsible Instructor	T. Keviczky	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course starts with the modelling of so-called networked control systems (NCS). Networked control systems are systems in which the communication between plant and controller takes place via a (e.g. wireless) network. Such network-based communication leads to imperfections in sensor and control signals, such as time-varying and uncertain sampling intervals and delays, packet dropouts, scheduling constraints, etc. The modelling framework introduced in the course is subsequently used to support stability analysis, using characterizations based on linear matrix inequalities (LMI). Such stability analysis allows to study trade-offs between requirements on the controller, the network and plant properties. The second part of the course deals with the aspect of the distributed control of networked systems. In particular, distributed optimization methods and various decomposition techniques (primal, dual, augmented Lagrangian / proximal point method, ADMM), links to consensus algorithms, and their application in networked multi-vehicle distributed robotics problems. Online optimization-based control approaches such as distributed model predictive control for multivehicle cooperation, distributed LQR and decomposition based methods that are applicable to collections of mobile agents. The methods will be illustrated on application examples including cooperative rendezvous, distributed formation control, spacecraft formation flight, and robotic networks.	
Study Goals	The student must be able to: 1. create models for networked control systems with network-induced uncertainties / effects / imperfections, including time-varying and uncertain sampling intervals, delays, packet dropouts, and scheduling constraints 2. analyse the stability of NCS (involving the above effects), e.g. by applying LMI-based stability characterizations 3. describe and apply decomposition techniques for distributed optimization problems 4. describe and apply consensus algorithms to multi-agent coordination problems 5. solve cooperative control problems by implementing a distributed model predictive control approach 6. analyse and evaluate the stability and convergence of distributed optimization and control methods 7. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback	
Education Method	Lectures	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course. Python is also allowed for some exercises/assignments.	
Literature and Study Materials	Course material: Lecture slides including additional reading material in the form of papers and textbooks are made available online.	
Assessment	1. This course is graded by four assessments, i.e., written assignment reports, each representing 25% of the final grade. 2. For some assignments part of the grading is done via Peer Review. 3. In order to complete the course, each assessment must be completed with at least a mark of 5,0 and the corresponding assignment report submitted before the deadline. In the event of a late submission, or a lower than 5,0 mark for any of the reports, no final mark will be given.	
Department	3mE Department Delft Center for Systems and Control	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Embedded Systems

Specialisation courses Control (choose at least 15 EC) 2022

CS4215	Quantitative Performance Evaluation for Computing Systems	5
Responsible Instructor	Dr. Y. Chen	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Today's computing systems become ever complex, due to the rapid development of hardware and software technology. It is challenging to design and run computing systems that guarantee users performance requirements in a resource efficient way. Various quantitative methods are applied to capture such complex system dynamics and predict metrics of interests, from the designing phase of the systems to the runtime performance, e.g., job response times and system anomaly. To optimize the performance of computing systems, a deep understanding on those methods and their applications on the system design are essential. Having practical hand-on experience on designing experiments, deriving models, and validating results with benchmark systems will prepare students to tackle challenges of real world computing systems.	
Study Goals	LO1. Design full/fractional factorial experiments for multi-variate regression analysis, e.g., finding critical parameters for deep learning clusters LO2. Apply queueing theory to analyse and predict run-time performance of applications, e.g., the average response times of on-line ML training service LO3. Apply machine learning models to analyse and predict the system dependability, e.g., root cause analysis for machine failure. LO4. Conduct experiments to profile applications and extract their workload parameters on real systems, e.g., e.g., deep learning clusters LO5. Develop resource management policies and validate them on real computing systems, e.g., deep learning clusters	
Education Method	Lectures: 7 weeks X 2-4h Practical: Derive, validate and evaluate performance models and resource management strategies for a chosen system via homework and group project. Multiple types of computing and network systems can be chosen from. Deliverables include a report and group presentation.	
Books	1) Performance Modeling and Design of Computer Systems: Queuing Theory in Action by Mor. Harchol-Balter 2) Design and Analysis of Experiments by Douglas C. Montgomery, 3) The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer Series in Statistics.	
Assessment	Homework (40%): 2 individual homework Group project (60%): group project report and presentation	
disclaimer: information may change depending on the developments around the coronavirus.		

RO47019	Intelligent Control Systems	4
Responsible Instructor	Dr. C. Della Santina	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Fundamentals of control theory	
Course Contents	Theory and application of intelligent control system i.e. the combination of machine learning and control theory. Topics: * Introduction to nonlinear control * Introduction to intelligent control: model learning, control learning * Fuzzy sets and systems * Artificial neural networks, deep learning (Reservoir Computing, CNN, Geometric Deep Learning) * Gaussian processes * Reinforcement learning * Imitation learning * Adaptive control * Iterative learning control * Examples of real-world applications	
Study Goals	At the end of this course, you will be able to: LO1 Explain the fundamentals of 'intelligent control' techniques - namely fuzzy logic, iterative control, Gaussian processes, artificial neural networks, etc. LO2 Implement 'intelligent control' techniques in MatLab and Simulink or Python. LO3 Reason on the expected performance of a given intelligent controller when applied to a given control problem. LO4 Propose an original, intelligent control scheme that can provide motor intelligence to a nonlinear and uncertain system (e.g., robot within a partially known environment).	
Education Method	Lectures and one assignment.	
Course Relations	Course title until 20/21 "Knowledge Based Control Systems" Course code until 21/22 "SC42050"	
Literature and Study Materials	Lecture notes: R. Babuska. Knowledge-Based Control Systems. Slides and other course material (video lectures, software, demos, notes) can be downloaded from BrightSpace.	
Assessment	The examination will be as follows: Practical assignment and written exam, open book. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Remote open book with several fraud prevention measures. In accordance with the master program regulations: * The students will have the opportunity for a resit if they fail the written examination. * The groups that are graded between 5 and 5.75 in an assignment, will have two weeks to integrate the assessors feedback. If this is done in a satisfactory way, they can be rewarded with a mark that is not higher than 6/10. The exact instruction for the addition will be announced on Brightspace.	
Tags	Artificial intelligence Complex Mathematics Group work Linear Algebra Mathematics Matlab Modelling	
Department	3mE Department Cognitive Robotics	
Contact	Cosimo Della Santina, C.DellaSantina@tudelft.nl	

SC42001	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. Due to COVID-19 the exam may be an online exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Department	3mE Department Delft Center for Systems and Control	

SC42015	Control Theory	6
Responsible Instructor	T. Keviczky	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students following this course are expected to have a strong background in the following topics: - Linear algebra (basic concepts up to generalized eigenvectors and Jordan canonical form) - Calculus (smoothness, Jacobian, gradient) - Differential equations (existence, uniqueness, Laplace transform) - Classical control (transfer functions, block diagram algebra, Bode/Nyquist, etc.)	
Course Contents	- State-space description of multivariable linear dynamic systems, interconnections, block diagrams - Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation - Dynamic response, relation to modes, the matrix exponential and the variation-of-constants formula - Realization of transfer matrix models by state space descriptions, coordinate changes, normal forms - Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback - LQ regulator, robustness properties, algebraic Riccati equations - Observability, detectability, unobservable modes, state-estimation observer design - Output feedback synthesis (one- and two-degrees of freedom) and separation principle - Disturbance and reference signal modeling, the internal model principle	
Study Goals	The student is able to apply the developed tools both to theoretical questions and to simulation-based controller design projects. More specifically, the student must be able to: - Translate differential equation models into state-space and transfer matrix descriptions - Linearize a system, determine equilibrium points and analyze local stability - Describe the effect of pole locations to the dynamic system response in time- and frequency-domain - Verify controllability, stabilizability, observability, detectability, minimality of realizations - Sketch the relevance of normal forms and their role for controller design and model reduction - Describe the procedure and purpose of pole-placement by state-feedback and apply it - Apply LQ optimal state-feedback control and analyze the controlled system - Reproduce how to solve Riccati equations and describe the solution properties - Explain the relevance of state estimation and build converging observers - Apply the separation principle for systematic 1dof and 2dof output-feedback controller design - Build disturbance and reference models and apply the internal model principle	
Education Method	Lectures and Exercise Sessions	
Computer Use	The exercises will be partially based on Matlab in order to train the use of modern computational tools for model-based control system design.	
Course Relations	This course is intended primarily for MSc students that are enrolled in the Systems & Control MSc program! For students who are interested in Systems & Control within their own MSc program, the SC42001 Control System Design course is recommended instead. There is a large overlap between the two courses in terms of topics, but SC42001 emphasizes broader concepts of control, it is more application-oriented, and does not discuss the technical proofs and mathematical background so extensively.	
Literature and Study Materials	B. Friedland, Control System Design: An Introduction to State-space Methods. Dover Publications, 2005 K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton and Oxford, 2009 http://www.cds.caltech.edu/~murray/amwiki/index.php?title=Main_Page	
Assessment	Written mid-term examination (15%) and written final examination (85%). For the resit examination there will be a written examination (100%) for which the mid-term result will not count. If on-campus exams will not be possible, then all exams will be organized online (via Ans).	
Department	3mE Department Delft Center for Systems and Control	

SC42025	Filtering & Identification	6
Responsible Instructor	Dr. M. Kok	
Instructor	Dr.ir. K. Batselier	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The objective of this course is to apply filtering and identification methods. The focus is both on deriving these methods and on using these methods on real-life data sequences. The course focuses on stochastic and recursive least squares, on (extended) Kalman filtering, on prediction error methods and on subspace identification. The course also includes a review of specific topics from linear algebra, dynamical system theory and statistics, that are relevant for filtering and system identification.	
Study Goals	At the end of the course you should be able to: 1) Apply filtering and identification methods to solve previously unseen estimation problems. As sub-learning goals you should be able to: a. Relate basic linear algebra, signal processing techniques and probability theory to physical properties of a (dynamical) system. b. Derive the solution of the stochastic and recursive least squares problems from both a Bayesian perspective and in terms of unbiased / minimum variance properties. c. Derive the Kalman filter from both a Bayesian perspective and in terms of unbiased / minimum variance properties. d. Derive subspace and prediction error identification methods. e. Motivate the steps that should be taken in the derivations in b-d. 2) Apply filtering and identification methods to estimate the state and identify the model using real-life data sequences. As sub-learning goals you should be able to: a. Implement a (stochastic / recursive) least squares problem, an (extended) Kalman filter and identification methods in Matlab. b. Apply and motivate practical aspects of filtering and identification.	
Education Method	Lectures 0/6/0/0	
Literature and Study Materials	Book Filtering and System Identification: A Least Squares Approach by Michel Verhaegen and Vincent Verdult. ISBN: 13-9780521875127	
Assessment	Written exam (open book) and practical exercises. If needed due to covid-19 restrictions, the exam will be online.	
Remarks	The software package Matlab is needed to solve the practical exercise.	
Department	3mE Department Delft Center for Systems and Control	

SC42045	Control Systems Lab	4
Responsible Instructor	Dr. R. Ferrari	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	Design, practical implementation and evaluation of a digital control system: - using mechatronic laboratory systems or high-fidelity dynamical models (inverted pendulum, 'helicopter' model, inverted wedge, 'acrobot' system) - applying both standard and advanced control methods (state-feedback, output-feedback, system identification, adaptive control) - using MATLAB/Simulink and the Real-Time toolbox In this course you will use a discrete-time approach, in which the system to be controlled is modeled both by discretizing an available continuous-time physical model and by using system identification. A systematic, MATLAB-supported design methodology is followed, using a state estimator (observer) and a state-feedback controller. Lectures ----- Lecture 1: Introduction. Course overview and goals. Description and mathematical models of the laboratory setups. System identification methods. Experiment design, model validation. Lecture 2: Modeling and identification. Mathematical modeling of physical systems, parameter estimation and tuning, model validation, simulation. Lecture 3: Design of digital controllers. Pole placement. Observers and output feedback. More control architectures. Lecture 4: Design of digital controllers. Recapitulation of computer-control design methodology. Design example and implementation in MATLAB / Simulink. Lab sessions ----- Laboratory sessions in times according to the students preference (within lab availability constraints). The scheduled lecture times can be used for consulting the approach and results with the lecturers. In case laboratory sessions cannot be held on site at the university using physical laboratory systems students will use dynamical models on their own computers. Remote sessions with the instructor and assistants will be organised.	
Study Goals	The course goals are to gain hands-on experience with the design and implementation of a computer-controlled system; and to understand computer-controlled systems in terms of design, analysis and implementation. In particular: - to understand discrete-time systems, - to be able to design sampled-data controllers, - to understand issues connected with implementation, - to be able to design and implement a controller for a simple physical process.	
Education Method	In the first two weeks, four lectures and a lab demo are given in order to refresh the theoretical and methodological background. Then, the students work in groups of three in the lab with a setup of their choice. If this is not possible due to access restrictions at the university, students will work at home using a high fidelity simulation. The assignment is stated in terms of the control objective and the mathematical model of the process to be controlled is provided. The results will be summarized in a report and a final presentation will be given. The grade is determined on the basis of the report and the presentation (i.e., there is no written exam).	
Assessment	- Group presentation: 30% of the grade - Final written report: 70% of the grade	
Remarks	Due to possible restrictions due to COVID-19 outbreak, the course may be given partly or fully online.	
Tags	Computer Engineering Design Embedded systems Group work Mathematics Matlab Mechatronics Modelling Practicals Programming Software Project Signals and Systems	
Department	3mE Department Delft Center for Systems and Control	

SC42056	Optimization for Systems and Control	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	basic experience with Matlab	
Course Contents	Essentially, almost all engineering problems are optimization problems. If a civil engineer designs a bridge, then one of the main objectives is to obtain the cheapest design or the design that can be implemented most rapidly, where of course several specifications and constraints such as size, strength, safety, etc. have to be taken into account. When developing a new type of engine, we look for the most economical design, the cheapest design, or the design with the highest performance. A process engineer wants a production unit to deliver a final product of maximal quality, with minimal expenditure of energy or with maximal output flow. When composing a portfolio, a financial engineer tries to maximize the expected profits, subject to the given risk constraints. So we encounter optimization problems in almost every engineering field. How can we solve such an optimization problem? That is the topic that will be addressed in this course. We will in particular discuss several classes of optimization problems and next focus on how to select most efficient numerical algorithms to solve a given optimization problem. The examples and case studies of this course are primarily oriented towards systems and control. More specifically, the following topics will be treated: * Linear Programming * Quadratic Programming * Nonlinear Optimization * Constraints in Nonlinear Optimization * Convex Optimization * Global Optimization * Matlab Optimization Toolbox * Multi-Objective Optimization * Integer Optimization	
Study Goals	After this course the students should be able to select the most efficient and best suited optimization algorithm for a given optimization problem. They should have insight into the basic operation of some of the most important and relevant optimization algorithms. They should be able to reduce the complexity of the problem using simplifications and/or reformulations so as to augment the efficiency of the solution approach. Finally, they should be able to solve the resulting optimization problem in practice.	
Education Method	lectures + 2 assignments Depending on the corona measures, the lectures will be taught on-campus or online.	
Assessment	Attending the lectures is not mandatory, provided the recorded lectures and/or lecture materials are the processed via self-study. exam (counts for 75% of the final marks) + 2 assignments (assessed through written reports, count for 25% of the final marks) Important: Partial marks for the exam or the assignments do not carry over from one academic year to the next. Depending on the of measures resulting from COVID-19, the prescribed exam will either be an open-book online exam, or if the opportunity arises for examinations on campus, a closed-book written exam (fully closed-book, no calculators nor formulas sheets allowed).	
Permitted Materials during Tests	none	
Department	3mE Department Delft Center for Systems and Control	
Contact	questions are preferably asked via the Discussion forum on Brightspace or (for questions on the material of the lectures) during the online or on-campus lectures	

SC42061	Nonlinear Systems Theory	3
Responsible Instructor	Dr.ing. S. Wahls	
Contact Hours / Week x/x/x/x	0.6.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	- Linear state space models (e.g. SC42015 Control Theory) - Basic knowledge of (multivariable) calculus, differential equations and linear algebra	
Course Contents	- Fundamentals of nonlinear systems (basic properties, solvability, types of equilibrium points) - Analysis of planar systems (graphical representations, periodic orbits, elementary bifurcations) - Stability theory for autonomous nonlinear systems (Lyapunov's methods, invariance principle) - Advanced stability theory (time-varying systems, input-to-state stability, passivity) - Observers (local, global and high-gain, extended Kalman filter)	
Study Goals	Students can: - determine if a system is nonlinear, time-invariant, autonomous, etc. - determine time intervals over which a system is solvable - sketch and interpret graphical representations of 2d systems - check for the existence or absence of periodic orbits in 2d systems, compute bifurcation points, and classify elementary bifurcations - determine if equilibrium points of autonomous nonlinear systems are (asymptotically, exponentially, ...) stable - determine if equilibrium points of certain classes of non-autonomous systems are (asymptotically, exponentially, ...) stable - construct observers for certain classes of nonlinear systems	
Education Method	Lectures, problem sets and instruction sessions.	
Literature and Study Materials	- H. K. Khalil: "Nonlinear Control", Pearson, 1st Global Edition, 2015. ISBN 9781292060507. - Additional handouts	
Assessment	Written exam, depending on the COVID-19 situation either on campus or online.	
Department	3mE Department Delft Center for Systems and Control	

SC42075	Modelling and Control of Hybrid Systems	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	core Systems & Control topics (including state-space models, discrete-time and continuous-time models, stability, linearization), optimization basics (linear programming), Matlab	
Course Contents	General introduction, examples of hybrid systems and motivation Modeling frameworks (automata, hybrid automata, piecewise-affine systems, complementarity systems, mixed logic dynamical systems, Petri nets) Properties and analysis of hybrid systems (well-posedness, Zeno behavior, stability, liveness, safety, ...) Control of hybrid systems (switching controllers, model predictive control) Verification and tools	
Study Goals	Recent technological innovations have caused a considerable interest in the study of dynamical processes of a mixed continuous and discrete nature. Such processes are called hybrid systems and are characterized by the interaction of time-continuous models (governed by differential or difference equations) on the one hand, and logic rules and discrete-event systems (described by, e.g., automata, finite state machines, etc.) on the other. A hybrid system also arises in practice when continuous physical processes are controlled via embedded software that intrinsically has a finite number of states only (e.g., on/off control). Recent interest in hybrid systems is stimulated by developments in nonlinear control theory, intelligent control, adaptive control, and computer science. The purpose of the course is to introduce a variety of hybrid systems modeling, analysis and control techniques.	
Education Method	lectures + assignment Depending on the corona measures, the lectures will be taught on-campus or online.	
Assessment	Attending the lectures is not mandatory, provided the recorded lectures and/or lecture materials are the processed via self-study. written exam (closed-book, no calculators, counts for 60% of the final marks) + assignment (assessed through written report, counts for 40% of the final marks) In case of unforeseen circumstances or measures resulting from COVID-19, the exam will be organized as an open-book take-home exam.	
Department	Important: partial marks for exam or assignment do not carry over from one academic year to the next 3mE Department Delft Center for Systems and Control	
Contact	Questions are preferably asked via the Discussion forum on Brightspace, or (for questions on the material of the lectures) during the online or on-campus lectures	

SC42095	Control Engineering	3
Responsible Instructor	A. Dabiri	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	SC42015, SC42000 or similar. Knowledge of classical control techniques (systematic and realistic PID design, frequency domain approaches) as well as state space theory is required.	
Course Contents	Computer control. Sampling of continuous-time signals. The sampling theorem. Aliasing. Discrete-time systems. State-space systems in discrete-time. The z-transform. Selection of sampling-rate. Analysis of discrete-time systems. Stability. Controllability, reachability and observability. Disturbance models. Reduction of effects of disturbances. Design methods. Approximations of continuous design. Digital PID-controller. State-space design methods. Observers. Pole-placement. Optimal design methods. Linear Quadratic control. Prediction. LQG-control. Implementation aspects of digital controllers.	
Study Goals	The student must be able to: 1. describe the essential differences between continuous time and discrete-time control 2. transform a continuous time description of a system into a discrete-time description 3. calculate input-output responses for discrete-time systems 4. analyse the system characteristics of discrete-time systems 5. employ a pole-placement method on a discrete-time system 6. implement an observer to calculate the states of a discrete time system 7. apply optimal control on discrete-time systems 8. describe the functioning of the Kalman-filter as a dynamic observer	
Education Method	Lectures and computer exercises	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course.	
Literature and Study Materials	Course material: Lecture notes are made available on Brightspace Textbooks: K.J. Astrom, B. Wittenmark 'Computer-controlled Systems', Prentice Hall ,1997, 3rd edition. L. Keviczky et. al 'Control Engineering', Springer, 2019. L. Keviczky et. al 'Control Engineering: MATLAB Exercises', Springer, 2019. References from literature: B.C. Kuo 'Digital Control Systems', Tokyo, Holt-Saunders, 1980 G.F. Franklin, J.D. Powell 'Digital Control of Dynamic Systems', 1989, 2nd edition, Addison-Wesley	
Assessment	Project assignment	
Remarks	Old course code: WB2305 The project assignment can be completed only during the quarter when the course is offered (i.e. the project has no resit during other periods).	
Department	3mE Department Delft Center for Systems and Control	

SC42100	Networked and Distributed Control Systems	3
Responsible Instructor	T. Keviczky	
Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course starts with the modelling of so-called networked control systems (NCS). Networked control systems are systems in which the communication between plant and controller takes place via a (e.g. wireless) network. Such network-based communication leads to imperfections in sensor and control signals, such as time-varying and uncertain sampling intervals and delays, packet dropouts, scheduling constraints, etc. The modelling framework introduced in the course is subsequently used to support stability analysis, using characterizations based on linear matrix inequalities (LMI). Such stability analysis allows to study trade-offs between requirements on the controller, the network and plant properties. The second part of the course deals with the aspect of the distributed control of networked systems. In particular, distributed optimization methods and various decomposition techniques (primal, dual, augmented Lagrangian / proximal point method, ADMM), links to consensus algorithms, and their application in networked multi-vehicle distributed robotics problems. Online optimization-based control approaches such as distributed model predictive control for multivehicle cooperation, distributed LQR and decomposition based methods that are applicable to collections of mobile agents. The methods will be illustrated on application examples including cooperative rendezvous, distributed formation control, spacecraft formation flight, and robotic networks.	
Study Goals	The student must be able to: 1. create models for networked control systems with network-induced uncertainties / effects / imperfections, including time-varying and uncertain sampling intervals, delays, packet dropouts, and scheduling constraints 2. analyse the stability of NCS (involving the above effects), e.g. by applying LMI-based stability characterizations 3. describe and apply decomposition techniques for distributed optimization problems 4. describe and apply consensus algorithms to multi-agent coordination problems 5. solve cooperative control problems by implementing a distributed model predictive control approach 6. analyse and evaluate the stability and convergence of distributed optimization and control methods 7. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback	
Education Method	Lectures	
Computer Use	Matlab/Simulink is used to carry out the exercises of this course. Python is also allowed for some exercises/assignments.	
Literature and Study Materials	Course material: Lecture slides including additional reading material in the form of papers and textbooks are made available online.	
Assessment	1. This course is graded by four assessments, i.e., written assignment reports, each representing 25% of the final grade. 2. For some assignments part of the grading is done via Peer Review. 3. In order to complete the course, each assessment must be completed with at least a mark of 5,0 and the corresponding assignment report submitted before the deadline. In the event of a late submission, or a lower than 5,0 mark for any of the reports, no final mark will be given.	
Department	3mE Department Delft Center for Systems and Control	

SC42110	Dynamic Programming and Stochastic Control	5
Responsible Instructor	Dr. P. Mohajerin Esfahani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Solid knowledge of undergraduate probability, especially conditional distributions and expectations, and Markov chains. Mathematical maturity and the ability to write down precise and rigorous arguments are also important. A class in analysis will be helpful, although this prerequisite will not be strictly enforced.	
Course Contents	The course covers the basic models and solution techniques for stochastic control problems and sequential decision-making under uncertainty. The course starts with a class of stochastic processes known as Markov chains. We then extend the model to include control variables (Markov Decision Processes, or in short MDPs) and study optimal control of a dynamical system over both a finite and an infinite number of stages. We also discuss model-free techniques (or reinforcement learning) as well as approximation schemes for problems involving large state spaces. Different applications from engineering and finance to operations research and management science will be covered.	
Study Goals	By the end of the course, the student must be able to - model real-world control and decision-making problems under uncertainty and in dynamic environments using the framework of discrete-time Markov decision process; - compute relevant technical concepts including hitting probabilities, expected hitting times, invariant distributions, and the long-run proportion of time spent in a given state; - formalize the optimal decision-making problem and (rigorously) derive the respective dynamic programming principle and useful properties of the desired solution; - use the dynamic programming principle and solve the problem using different computational approaches; - explore the technical literature in control and operations research, and undertake independent self-study and research in the future.	
Education Method	Lectures	
Books	- Dynamic Programming and Optimal Control, 3rd edition, D. Bertsekas, Athena Scientific, 2005 - Introduction to Probability, D. Bertsekas and J. Tsitsiklis, Athena Scientific, 2002	
Assessment	Written exam (depending on the corona situation the exam may be online.)	
Department	3mE Department Delft Center for Systems and Control	

SC42125	Model Predictive Control	4
Responsible Instructor	Dr.ing. S. Grammatico	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Model Predictive Control (MPC) is perhaps the most effective optimal control strategy for constrained dynamical systems. The basic concept of MPC is to exploit a dynamic model to forecast the system behavior, and optimize the forecast to determine the control input as the best decision at the current time. With emphasis on constrained linear systems, the course will present the theoretical fundamentals of MPC, such as Lyapunov stability, optimality and robustness, and its computational methods, such as quadratic programming.	
Study Goals	- Derive state prediction matrices from discrete-time linear models - Design the cost function, state and input constraints - Design MPC controllers with guaranteed recursive feasibility and asymptotic stability via appropriate terminal cost and constraint set - Formulate and solve constrained-linear-quadratic MPC problems via quadratic programming - Implement and simulate closed-loop systems controlled by MPC on Matlab - Design MPC controllers with integral action for reference tracking	
Education Method	Lectures, instructions.	
Assessment	Final project and oral exam.	
Elective	Yes	
Tags	Mathematics Matlab Optimisation	
Department	3mE Department Delft Center for Systems and Control	

SC42130	Fault Diagnosis and Fault Tolerant Control	4
Responsible Instructor	Dr. R. Ferrari	
Contact Hours / Week x/x/x/x	0.0.0.4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Control Systems Design, System Identification, Signal Processing, Statistics and Probability	
Summary	In critical systems such as transportation systems, power plants, distribution networks or life-support systems it is not enough to design a control system that meets the required performance levels. It is also of paramount importance to guarantee that such requirements, or a subset of them, are met also during non ideal conditions such as in the presence of physical failures or cyber attacks. Most of all, safety must be guaranteed at all times. This can be obtained by designing into the control system additional functionalities that will allow to detect the occurrence of faults or cyber attacks, and deploy adequate countermeasures, which may include the reconfiguration of the control system.	
Course Contents	Faults and failures in dynamical systems; Components and Services model; Availability and Reliability; FTA (Fault tree Analysis); FMEA (Failure Modes and Effects Analysis); Signal based methods for Diagnosis; Model based methods for Diagnosis; Residuals computation and evaluation; Learning of fault model; Fault Tolerant Control (Model Matching, Virtual Sensor and Actuator, Adaptive Control)	
Study Goals	after the course the student should be able to: 1) analyse the structure of a system and identify main components, the services provided by each and carry out a Fault Modes and Effects Analysis; 2) analyse and model faults or cyber attacks that can occur in a given dynamical system; 3) design an algorithm for detecting, isolate and identify them; 4) design a policy for reconfiguring the control system in order to accommodate them.	
Education Method	Flipped classroom: video lectures online, in-class exercises, homework	
Literature and Study Materials	Recorded lectures, Slides, Example exercises, Suggested textbook: Blanke, M., Kinnaert, M., Lunze, J., Staroswiecki, M., & Schröder, J. (2006). Diagnosis and fault-tolerant control (Vol. 691). Berlin: Springer.	
Assessment	Homework 30%, final exam (written) 70%	
Elective	Yes	
Tags	Mathematics Matlab Mechatronics Modelling Numeric Methods Process Signals and Systems	
Department	3mE Department Delft Center for Systems and Control	

SC42145	Robust Control	3
Responsible Instructor	Prof.dr.ir. J.W. van Wingerden	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	· Recap on background in linear systems theory and classical feedback control · Multivariable system control: Nyquist, interaction, decoupling · Directionality in multiloop control, gain and interaction measure · Stabilizing controllers and the concept of the generalized plant · Parametric uncertainty descriptions, approximations · The general framework of robust control · Robust stability analysis · Nominal and robust performance analysis · The H-infinity control problem · The structured singular value: Definition of μ · μ synthesis, DK-iteration, role of uncertainty structure. · Design of robust controllers, choice of performance criterion and weights	
Study Goals	The student is able to reproduce theory and apply computational tools for robust controller analysis and synthesis. More specifically, the student must be able to: 1. substantiate relation between frequency-domain and state-space description of dynamical systems 2. define stability and performance for multivariable linear time-invariant systems 3. construct generalized plant for complex system interconnections 4. describe parametric and dynamic uncertainties 5. translate concrete controller synthesis problem into abstract framework of robust control 6. reproduce definition of the structured singular value 7. master application of structure singular value for robust stability and performance analysis 8. design robust controllers on the basis of the H-infinity control algorithm 9. apply controller-scalings iteration for robust controller synthesis	
Education Method	Lectures + Assisted lectures+Computer sessions	
Literature and Study Materials	S. Skogestad, I. Postlethwaite, Multivariable Feedback Control, 2nd Edition. John Wiley & Sons, 2005. References from literature: K. Zhou, J.C. Doyle, K. Glover, Robust and optimal control, Prentice Hall, 1996 D.-W. Gu, P.Hr. Petkov, M.M. Konstantinov, Robust Control Design with Matlab. Springer Verlag, London, UK, 2005	
Assessment	Controller design exercise (report and code)	
Remarks	Old course code: SC42010	
Department	3mE Department Delft Center for Systems and Control	

SC42150	Statistical Signal Processing	3
Responsible Instructor	C.S. Smith	
Responsible Instructor	Dr. P. Mohajerin Esfahani	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	1. Introduction concepts a. Dynamic models, measurement models (probabilistic form) with examples b. Introduce concepts estimation, filtering, smoothing and identification 2. Stochastic signals a. What is the difference between a RV and a RP? b. RPs i. White noise ii. Radom walk iii. Poisson iv. Binary v. Gaussian vi. Autoregressive/moving average c. Wide sense stationary d. Auto/cross correlation and covariance e. Power/cross spectral density 3. Linear filtering a. FIR b. IIR c. Causal-IIR d. Examples i. Noise cancelation ii. Deconvolution iii. One-step a head prediction	
Study Goals	After the course, students can - describe and analyze stochastic signals in the time and frequency domain and can use these insights to quantify noise distorted measurement signals; - use the concepts such as auto and cross correlation as well as auto and cross signal spectra; and estimate these quantities from finite series of measurements. - calculating the change in the correlation functions and spectra when stochastic signals are filtered by a linear dynamic system; - solving the inverse problem, i.e. how they can calculate the formative filter from the stochastic spectra. - estimate the information-carrying components from disturbed measuring signals, - derive and use the linear optimal Wiener filter and Least Squares estimators to solve noise reduction problems. - describe and analyze a measurement signals in the time domain and can use these insights to derive a minimum variance estimate take incorporates a-priori statistical information.	
Education Method	Lecture with tutorial consisting of theoretical and python exercises	
Assessment	Written (online) exam and homework. The final mark is a weighted average of the mark obtained on the examination and that on the compulsory homework / python assignments.	
Department	3mE Department Delft Center for Systems and Control	

SC42155	Modelling of Dynamical Systems	3
Responsible Instructor	Dr. M. Mazo Espinosa	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Modelling. The course will cover the basics of Lagrangian modelling, I.e. employing energy balances and the Euler-Lagrange equations to arrive to closed form state-space dynamical equations for system analysis and control design. Bond-graph modelling will be covered as a powerful tool for modelling engineering systems, especially when different physical domains are involved. The basic ideas of model reduction, a technique to reduce the complexity of dynamical models, is also covered in the course. The fundamental differences and connections between continuous vs discrete time modelling will be discussed. Similarly, frequency domain modelling is also discussed with practical motivations for frequency vs time domain modelling. Finite State Models. A brief introduction to models traditionally employed for computation, e.g. finite state machines, automata, and petri-nets, will be provided. Particular attention will be given to the applicability of these models to provide high-level dynamic descriptions of discrete-event systems and for behaviour specifications of designs. Hybrid systems. The course will finalize with a short introduction to hybrid systems through the formalisms of: Hybrid automata and jump-flow systems, and presenting a brief number of interesting subclasses of hybrid systems, e.g. PWA systems, and timed -automata.	
Study Goals	The purpose of the course is to introduce the students to basic concepts and results in physical modeling and the theory of nonlinear control systems. After successful completion of the course, the student is able to - construct models of systems from the knowledge of physics on the basic laws of physics - construct models of systems using the theory of bond graphs. - construct models of systems using the Euler-Lagrange method - perform a model reduction for linear systems using a balanced realization - construct finite state models of simple computing and scheduling systems - analyze basic properties of finite state systems: blocking, determinism, and language expressed by a model. - describe cyber-physical systems as hybrid dynamical models. - select the appropriate modelling approach for a given problem.	
Education Method	Lectures 0/4/0/0	
Literature and Study Materials	Lecture notes and additional reader	
Assessment	2 Assignments (25% each) and a mini-project (50%)	
Department	3mE Department Delft Center for Systems and Control	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Embedded Systems

Thesis Project 2022

ES5000	Final Project	45
Responsible Instructor	Dr.ir. A.J. van Genderen	
Contact Hours / Week x/x/x/x	x/x/x/	
Education Period	1 2 3 4 Summer Holidays	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	ES5000 Final Project is the final part of the Master's degree programme. During this project, you will be required to demonstrate your ability to solve a research or engineering problem. The project must be carried out using the techniques of project management. You will begin by making a project plan in cooperation with your Masters thesis advisor. Several aspects of the project are defined within the plan, including the assignment, the frequency of interaction with the advisors, the milestones of the project and the resources and facilities offered by the faculty. You will be required to adhere to your plan throughout the project. It is obviously possible to adjust your plan under certain circumstances and after discussion with your daily supervisor. It is possible to do the project in industry. In that case it is important to note that you should not make arrangements with any company without prior consultation with your master coordinator and without having a TU Delft supervisor. At the end of the project, you will submit your Masters thesis, which must be written in English, and make an oral presentation of your work to the Thesis Committee. The Thesis Committee will announce the final mark, which is based on the quality of work, project performance, the thesis, the presentation and the subsequent discussion. More information about the graduation process: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-msc/.	
Study Goals	1. The student is able to design a research or engineering project: - The student is able to orientate on an interesting scientific or engineering question by means of a scientific literature study; - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research or engineering project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report about the project: - The student is able to write a research report that shows sufficient coherence of content; - The students is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research or engineering project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Supervised research, individual project	
Prerequisites	The student must have completed at least 60 EC of the Master's degree programme and be in possession of a Declaration of Credits Obtained (can be requested from EEMCS service desk) that shows that this is true. To be able to get a Declaration of Credits Obtained, the individual exam program (IEP) should have been approved by the Board of Examiners. Note: In some cases, the thesis supervisor may impose additional conditions for starting this project.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-msc/assessment/	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Embedded Systems

Suggestion free elective space: Internship & Projects 2022

EE5010	Internship 10-15 ECTS	10
Responsible Instructor	Prof.dr.ir. P.M. Sarro	
Course Coordinator	M.J.W. Smits	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Master students of Electrical Engineering, Computer Engineering or Embedded Systems can follow EE5010 as part of their free electives. The internship period should be from 8 weeks (for which you will receive 12 credits) up to three months (for which you will receive 15 credits). Internships longer than 3 months will NOT be approved! The internship should be of sufficient level to be part of your MSc curriculum performed at a company or institute, either in the Netherlands or abroad. An internship in a university outside the Netherlands can be accepted. An internship at TU Delft will not be accepted. When carrying out Internship EE5010 at a company or institute, the following conditions apply: The student participates in the company or institute culture in general, including sharing the company-established balance between on-site and online work. The student takes part in the day-to-day running of the company or institute similar to regular employees. This includes having regular consultation and interaction with colleagues and manager(s), and use of and access to the company-specific resources and tools applicable to the internship. Such resources and tools can be physical/on-site (e.g. measurement equipment, company machines and infrastructure) as well as virtual (online, software tools, databases, ...).	
Study Goals	There are three types of educational goals for the Internship: I. Profession related aspects Insight into the profession of the electrical engineer Insight into the duties and the responsibility of the engineer in a company II. Social-psychological aspects Learning to operate in another culture (both within and outside the company) Gaining insight into the position of the company with respect to comparable companies III. Professional knowledge and skills related aspects Learning to apply the gained knowledge and skills in a situation different from the (own) university situation Learning to acquire new knowledge and skills that are necessary to meet the requirements for the Internship	
Education Method	Detailed information on projects, procedures and application is available on the Brightspace course EE5010. The internship application form should be submitted to the Internship Office prior (!) to starting Internship EE. This application form contains a project description that the student writes with input from the company supervisor and a TU Delft supervisor. A completed internship application form, submitted to the EEMCS internship office, indicates that you can do the internship for study credit	
Literature and Study Materials	Internship regulations: programme Electrical Engineering, Embedded Systems and Computer Engineering. Enroll in Brightspace course EE5010 for more information about the Internship. You can find the step-by-step guide explaining all procedures, forms and internship offers. For additional information e.g. about grants and funding for internships abroad enroll in Brightspace course Study Abroad EEMCS.	
Prerequisites	In order to ensure that students start the projects with a solid skill set, students can apply for the Internship after the third quarter of the first year of the MSc EE, CE and ES programme. In order to start the Internship the following requirements must be fulfilled: 1. The Individual Exam Programme (IEP) must have been submitted to and approved by the Board of Examiners. 2. You have obtained 45EC before the start of your Internship. Detailed information on projects, procedures and application is available on the Brightspace course EE5010. The internship application form should be submitted to the Internship Office prior (!) to starting Internship.	
Assessment	Pass/Fail (no grade) For the assessment you need to submit: The company assessment form The TU Delft assessment form An internship and evaluation report. You can find guidelines for the report on Brightspace. In order to give the company supervisor sufficient time to read and check the report, yet to ensure that students receive timely feedback and final grade, your company supervisor is asked to assess your internship at the end of your internship period, within two weeks after submission of the reports, and send his assessment form to the TU Delft supervisor. Your TU Delft supervisor needs to assess your internship within two weeks after receiving the assessment of the company supervisor. The TU Delft supervisor will then send the assessment form (including the company supervisor assessment form) to the EEMCS Internship Office; internship-eemcs@tudelft.nl. The assessment form can be found on Brightspace EE5010. Step by step: - During your internship period you write the scientific report. - At the end of your internship, you have up to one week to finalize your evaluation report. - At the end of your internship, you send the scientific report to your company supervisor. - The company supervisor sends his assessment within 2 weeks after receiving the scientific report to the TU Delft supervisor. - You send both reports to the internship office. - The TU Delft supervisor sends his assessment within 2 weeks after receiving the report and the company assessment and send both assessments to the internship office. The assessment is based on the Company supervisor Assessment Form, the TU Delft supervisor Assessment Form and the Internship Report and will be graded with a pass or fail. Your mark will be processed as soon as the Internship Office EEMCS has received:	

Two digital assessment forms signed by the supervisors and send by the TU Delft supervisor to the internship office (you need to provide your supervisors with the assessment forms yourself. You can find them on Brightspace EE5010).
A digital copy of both reports sent by you to the internship office

ET4399	Extra Project (max. 15EC)	15
Responsible Instructor	Dr.ir. N.P. van der Meijs	
Responsible Instructor	Dr. R.A.C.M.M. van Swaaij	
Course Coordinator	Dr.ir. R.C. Hendriks	
Course Coordinator	M. Babaie	
Course Coordinator	S.M. Alavi	
Course Coordinator	Dr. D. Cavallo	
Course Coordinator	Dr.ir. A.J. van Genderen	
Course Coordinator	M.A. Kitsak	
Course Coordinator	Dr. A. Lekic	
Course Coordinator	J. Dong	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	This Extra Project offers students the opportunity to enhance lab skills (such as a specific training), academic skills (such as writing a paper for publishing or participating in a conference with a posterpresentation or paper) or for studying special topics (such as capita selecta with (guest) lecturers). Note: the Extra Project can only be taken as part of the free electives. The supervisor needs to be part of the scientific staff within the Faculty of Electrical Engineering, Mathematics and Computer Science, this means they need to be a Full Professor, Assistant Professor or Associate Professor. For the MSc SET programme they should be on the Green list.	
Study Goals	To be formulated by the student in the Extra Project proposal (see education method) and to be approved beforehand by the supervisor and the MSc coordinator.	
Education Method	Before the start of the Extra Project, a proposal of the project needs to be provided according to a specified format; the format of this Extra Project proposal can be downloaded from the Brightspace page of the Extra Project. The following information is required: name of the TUD supervisor, proposed number of credits, a motivation for the extra project, learning goals and a description of the planned activities and output. This project proposal needs to be approved by the TUD supervisor, and for EE and CE the MSc coordinator, and for SET the director of studies. When submitting the Individual Exam Programme to the Board of Examiners, the Extra Project proposal needs to be attached. After the assignment, a final report needs to be produced. The Board of Examiners may ask to see the report to check the study load and quality at the time of the final MSc exam. At most one Extra Project may be performed and it can only be taken in the free elective space of the MSc programme.	
Assessment	The Extra Project and the resulting report is evaluated and graded by the supervisor. The grade is Pass/Fail. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	

IFEEMCS520200	Interdisciplinary Advanced Artificial Intelligence Project	15
Responsible Instructor	Dr. P.K. Murukannaiah	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	This course is aimed at students who have already obtained a 30EC computer science/artificial intelligence background. Having a solid shared pool of knowledge on AI enables the students embarking on this course to work in a team on an AI challenge at a master level. Prospective students, for example, have completed an AI minor, or have a good amount of AI knowledge from their bachelor (e.g., Mechanical Engineering).	
Course Contents	Artificial Intelligence (AI) is an interdisciplinary challenge. This course aims to provide students with a unique experience to learn, develop and apply AI to a real-world challenge, working in an interdisciplinary project team. During the course, the project team executes AI research, and/or produces AI solutions, that address real-world problems. Students collaborate in a group of 4-6 students, ideally, of diverse backgrounds (including Computer Science). Mini-tutorials will be organized during the course to provide students with knowledge and skill-based input required to successfully complete their project. The topics of the mini-tutorials will be based on the initiative from the project supervisors and other AI researchers, e.g., from the TU Delft AI Labs. The course will include a kick-off, mini-tutorials, and lab sessions with project supervisors.	
Study Goals	Upon finishing this course, the student should be able to: 1. position an AI challenge in the broader literature, e.g., by performing a literature review on the AI challenge in a domain; 2. set-up a research strategy for an interdisciplinary AI project; 3. develop and deploy AI solutions according to a research strategy; 4. effectively collaborate with the team and supervisor; 5. evaluate and reflect on their research / developed AI solution; 6. assess the impact of deploying AI-based solutions and interventions on individuals, organizations, and society; and 7. report and present their research on a scientific level.	
Education Method	The students may work and discuss in groups. Each student is responsible for gathering, sharing and implementing information within the overall project. This allows for collaboration and communication as well as self-directed learning. At least once a week, the students meet with their supervisor. Peer feedback as well as feedback from the supervisor is organized on essential steps in the research project, which include: 1. Project proposal 2. Mid-term review, and 3. A scientific report, describing the problem, methods, results, and conclusion 4. A scientific presentation The mini-tutorials are organized as electives, i.e., students identify which tutorials contribute to their assigned task and field of expertise within the group project.	
Assessment	The scientific report and the project presentation are assessed by the project supervisor and an external examiner. The assessment based on the clarity of presentation, the soundness of the methods used, and the conclusions drawn. Resit next quarter: A repair option is allowed only for students that received a fail mark (<5.8). Overall, the resit mark will be capped to 5.8.	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

Year	2022/2023
Organization	Electrical Engineering, Mathematics and Computer Science
Education	Master Embedded Systems

Language courses 2022

TPM018A	English Grammar for the University	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The course is designed for students who want to improve their English grammar in order to communicate more effectively. We use a university grammar that addresses the limitations many students face when interacting in English. The focus of this course is on improving your grammar range and accuracy to help you express complex thoughts more clearly.	
Study Goals	During this course, students will: 1. expand their basic understanding of sentence structure in academic discourse. 2. demonstrate an understanding of more complex grammatical structures 3. write a variety of sentence types 4. practise the grammar needed to write an academic text 5. demonstrate an understanding of grammar through text assignments 6. write paragraphs which use targeted grammar structures	
Education Method	It is a 7-week course with a weekly 90-minute class. If registration is lower than expected, the course may be offered as self-study in combination with a weekly consultation hour.	
Literature and Study Materials	Grammar for Academic Purposes 2 - Steve Marshall This textbook is available in e-format only (approx. 20) or a combined e-book and paper book (approx. 30). Details of how to obtain a copy will be given after enrolment.	
Assessment	Weekly assignments with set deadlines. If deadlines are not met, students may be asked to leave the course.	
Elective	Yes	
Tags	English proficiency	
Category	BSc and MSc level	

TPM302B	Spoken English for Academic Purposes Advanced	2
Module Manager	Drs. M.A. Swennen	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This speaking course focuses on expanding and improving the language students need to participate at the university. A particular focus point of the course is the vocabulary needed at a higher level of English, both in an academic and a more general context. A range of sources for language input is offered, such as podcasts, videos, websites, or (academic) articles. The sessions revolve around student interaction.	
Study Goals	The first aim of the course is to improve and expand the language skills needed at a higher level of English. These include paraphrasing and sharing information with others, using polite language, giving feedback, and presenting information. Another study course goal is to expand and improve the students' general vocabulary.	
Education Method	Group and pair work, combined with (individual) assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Literature and Study Materials	To be announced in Brightspace.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English language skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for either of the courses offered at that level.	
Assessment	For this course you need a recommendation to take courses at the Advanced level. Class participation and weekly assignments. An 80% attendance rate is obligatory, which means you may only miss one session. To keep your place in the group you must attend the first session. The assignments must be submitted each week before the deadline set by the instructor.	
Enrolment / Application	You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class after session 2. Before enrolling in a group you must first take our Placement Test. This is not an actual test but a tool to ensure that you are taking the course that is the most suitable for your level of English. For more information on this 'test' please visit our website: www.itav.tudelft.nl Enrolment for the course (and the Placement Test) is via Brightspace. You do not need to register for this course in Osiris. Please note that you must also choose a group to reserve a place in the class. Type 'TPM302B' in Course search (Catalog). Please make sure you select the correct academic year and quarter. Register in the course AND in a group that suits your schedule.	
Remarks	The course will only be offered if there is a sufficient number of participants. Please note that it is not a presentation course.	
Elective	Yes	
Tags	English proficiency Group work Small groups	
Category	BSc and MSc level	

TPM303A	Intermediate Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on the acquisition of necessary language skills in the areas of grammar and usage, as well as reading and writing at a level appropriate to university study. Students will develop skills that develop an idea with relevant support for that idea by structuring the text according to current writing conventions, using various transitions to link ideas and demonstrating a knowledge of grammar and punctuation. Students will edit their own texts and work with other students in peer-editing groups.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who would like to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at the Intermediate level.	
Books	Advance in Academic Writing 1 - Steve Marshall This textbook is available in e-format (approx. 34) or a combined e-book and paper book (approx. 55). Details of how to obtain a copy will be given on Brightspace after enrolment.	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at intermediate level. After receiving your Placement Test results you can enrol via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q2 and Q4. However, the course will only be offered if there are sufficient participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM304A	Advanced Writing in English for the University	2
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This writing course is a 7 week course with 1 lesson a week (2 hours) plus approx. 6 hours of homework assignments per week. The course focuses on writing effective, well-structured and coherent academic texts in English. It looks at the writing process, including planning and structuring the text. Students will then develop paragraphs into longer academic texts and work on text coherence and cohesion. The course also pays attention to advanced academic vocabulary, sentence structure, punctuation and grammar.	
Study Goals	The aim of the course is to develop language skills and writing skills needed for producing academic texts. This course will give the student an understanding of the conventions of academic writing in English and some of the grammatical structures common in academic texts.	
Education Method	Individual, pair and group work, combined with assignments to do at home. Feedback on course components is given by the course lecturer and other students.	
Practical Guide	This is a course for both BSc and MSc students who already have a good understanding of academic writing, but who wish to expand and improve their English writing skills. When you have taken our Placement Test you will be recommended a language level. With this recommendation you can sign up for any of the courses offered at that level. For this course you need a recommendation to take courses at Advanced level.	
Books	Advance in Academic Writing 2 - Steve Marshall This textbook is available in e-format (approx. 34) or a combined e-book and paper book (approx. 55). Details of how to obtain a copy are given on Brightspace.	
Assessment	Students will be given a final writing assignment. In borderline cases teachers will take the final decision, based on the student's class participation and homework assignments.	
Enrolment / Application	Before enrolling in a group you must first take our Placement Test. For more information on this test please visit our website: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test For this course you need a recommendation to take courses at Advanced level. After receiving your Placement Test results you can enroll via Brightspace. Please note that you must also choose a group to reserve a place in the class. Once you have enrolled in the course, go to collaboration and select the group that suits your schedule. You do not need to register in Osiris for this course.	
Special Information	Note that you are required to spend an additional 6 hours (approx.) per week doing homework assignments. Please make sure that your schedule allows you this time.	
Remarks	An 80% attendance rate is obligatory, which means you may miss one session. To keep your place in the group you must attend the first session. You can drop the course with no penalty before the second session. Be aware of your "commitment" from the start, as you will still receive a grade [NV: Niet Verschenen--Not Attending], in Osiris even if you stop coming to class. The course is offered in Q1 and Q3. However, the course will only be offered if there is a sufficient number of participants.	
Elective	Yes	
Tags	English proficiency	
Targetgroup	This course is available to BSc and MSc students	

TPM305A	Writing a Masters Thesis in English	2
Module Manager	Drs. D.W. Laponder	
Instructor	S.F. Johnson	
Instructor	Drs. D.W. Laponder	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. D.W. Laponder	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This course is designed for students who have started their MSc thesis or are about to start on their graduation project. The main focus is on the writing process, which means that matters such as grammar, academic style and precision of formulation are considered at length. Particular attention is also paid to the importance of structure and organising ideas. Aspects of the writing process will be backed up by weekly reading and writing assignments.	
Study Goals	The course is primarily designed for students working on their Masters thesis. As the name of the course indicates, the focus throughout is on writing. Samples of your academic writing will be corrected in considerable detail and you will be taught how to structure an essay, report or thesis. Plenty of tips will be given on ways of perfecting your English and invigorating your writing. Your systematic mistakes will be pointed out to you so that you can become critical about your own writing and go on to do your own editing.	
Education Method	Learning by doing and learning from lectures, corrections, peer review and tutor feedback.	
Books	Academic Writing for Graduate Students, Essential Tasks and Skills ,3rd edition by John M. Swales & Christine B. Feak. ISBN: 978-0-472-03475-8	
Assessment	Continuous assessment. Participants are expected to submit weekly writing assignments on a subject that is closely related to their field of study. An 85% attendance record is required. This means you may miss not more than one session (preferably none). If you fall behind with your coursework or cannot attend a minimum of six classes, you may be asked to leave the course. You will receive the course credits when all conditions are met.	
Elective	Yes	
Tags	English proficiency	
Category	MSc level	

WM1115TU	Dutch Elementary 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch English	
Required for	After this course, participants can continue learning in Dutch Elementary 2 (course code WM1116TU)	
Expected prior knowledge	No prior knowledge on the Dutch language needed. Basic knowledge of English is required. If you already know about 1000 words in Dutch or more, we advise you to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 1 is a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A1.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- There is a two-step-enrolment procedure. 1: Enrolment in Preliminary program for Dutch Elementary 1 course This programme takes place one week before the actual course. It is meant for receiving information about the learning method. On top of that, you will study the first two texts and sit the introductory test. If you pass, you can enrol for the actual course Dutch Elementary 1. 2: Enrolment in the actual course Dutch Elementary 1 (WM1115TU) You can enrol for one of the groups. --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse Small groups	
Targetgroup	Students and others who want to learn basic Dutch and who have little or no prior knowledge of Dutch. People who already know some Dutch (approx. 1000 words or more) can contact delftsemethode@tudelft.nl for a level test.	
Self Test	If you want to do self-study, you can register for a written test + speaking test by sending an email to delftsemethode@tudelft.nl	
Category	MSc level	

WM1116TU	Dutch Elementary 2	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 1 (course code WM1117TU)	
Expected prior knowledge	A completed course Elementary 1 (WM1115TU). It is also possible to enrol if you have a similar level: CEF A2. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Elementary 2 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Bondi Sciarone e.a., "Nederlands voor buitenlanders. De Delftse Methode". Text book. 5th revised edition.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration.	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretarys office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1117TU	Dutch Intermediate 1	3
Module Manager	Drs. G. Hoezen	
Responsible for assignments	Drs. G. Hoezen	
Co-responsible for assignments	Dr. S.J. van Boxtel	
Contact Hours / Week x/x/x/x	3/3/3/3	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	Dutch	
Required for	After this course, participants can continue learning in Dutch Intermediate 2. This is a paid course, also for students and will not give you ECTS credits.	
Expected prior knowledge	A completed course Elementary 2 (WM1116TU). It is also possible to enrol if you have a similar level: CEF A2+. In that case you have to sit a level test. Contact delftsemethode@tudelft.nl .	
Parts	Preparing lessons at home by reading, listening and repeating Dutch texts, speaking during conversation class. Continual testing through listening tests, writing assignments and gap filling tests.	
Course Contents	Dutch Intermediate 1 is the second part of a basic Dutch language course aimed at every day communication. You will learn the language through reading, analysing and listening to texts on various topics as preparation for the lessons. The lessons themselves will be mainly devoted to speaking. Regular testing is also part of this course.	
Study Goals	The goal of this course is to acquire basic Dutch language skills on CEF level A2.	
Education Method	Interactive group sessions, individual work through online course materials.	
Literature and Study Materials	Wesdijk en Blom Sciarone e.a., "Tweede Ronde, herziene editie. Nederlands voor buitenlanders. De Delftse Methode". Text book.	
Assessment	The final grade is compiled of grades for speaking skills during class and several tests and assignments during the course period: writing assignments, listening tests and written gap filling tests. Attendance also contributes to the final grade. If this final grade is not sufficient, students sit a speaking test, a written test or a combination of both. These tests take place after the regular course period.	
Permitted Materials during Tests	No materials admitted during tests.	
Enrolment / Application	--- for MSc/BSc students --- Enrolment through Brightspace in the course Dutch Elementary 2 (WM1116TU) --- for others (PhDs, employees) --- Go to the website www.dm.tudelft.nl for registration	
Special Information	Brightspace, www.dm.tudelft.nl or delftsemethode@tudelft.nl	
Remarks	Regular courses take place at the campus in groups of max. 15 participants. We will also offer some groups in an online format. These groups have max. 9 participants. Check the website or Brightspace to see the current situation. There is an option to take this course as a self-study programme with a final test. Contact the secretaries office for registration. delftsemethode@tudelft.nl More information: www.dm.tudelft.nl	
Elective	Yes	
Tags	Broad Diverse	
Targetgroup	Students and others who want to improve their Dutch knowledge. If you are not sure whether is course is on the right level for you, please contact delftsemethode@tudelft.nl for a level test.	
Category	MSc level	

WM1135TU	Advanced English for the University	3
Module Manager	Drs. K.I. van der Linden	
Instructor	S.F. Johnson	
Instructor	Drs. K.I. van der Linden	
Instructor	Drs. M.A. Swennen	
Responsible for assignments	Drs. K.I. van der Linden	
Co-responsible for assignments	S.F. Johnson	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	This course is given at C1 CEFR (Common European Framework of Reference) level. Advanced English for the University is the highest level integrated skills course offered by the English Unit within the ITAV institute at TPM. It is designed for those who already have a thorough working knowledge of English grammar and vocabulary but who still wish to further perfect their proficiency in order to become more adept in both writing and speaking. Time will be devoted to: Academic texts and writing conventions Academic vocabulary and collocations Grammar and punctuation Presentations and discussions Language learning strategies Advanced English for the University is a 14-week course with 1 lesson a week (2 hours) plus approx. 4 hours of homework assignments per week. The groups tend to be small (max. 18 participants) and the sessions are highly interactive. Attendance is mandatory. Students can miss a maximum of two lessons. Courses begin in the first week of September and in the first or second week of February . Please see the website for Placement Test dates and for any further course details: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit	
Study Goals	By the end of the course students should be familiar with different types of academic texts academic writing conventions academic vocabulary with a specific focus on collocations common grammatical errors punctuation conventions structural devices to improve coherence various language resources such as online dictionaries	
Education Method	Students will do in-class exercises completed either in small groups or in pairs. They will also do a group PowerPoint presentation and a discussion led in pairs. The feedback on course components will be given both by the lecturer and by fellow course participants.	
Literature and Study Materials	Course book: Cambridge Academic English - An integrated skills course for EAP (Student's Book; C1 Advanced). Authors: Martin Hewings and Craig Thaine Publisher: Cambridge University Press, 2012 ISBN: 978-0-521-16521-1	
Prerequisites	To qualify for the course students need to have received an Advanced English for the University recommendation following the completion of the Placement Test. Please see in this connection: https://www.tudelft.nl/tbm/over-de-faculteit/afdelingen/stafafdelingen/itav/onderwijs/english-unit/placement-test	
Assessment	There is no final exam. Assessment is based on the written assignments and on the in-class discussions and presentations. The six writing assignments each account for 10% of your final grade while your grade for the discussion and your grade for the presentation each accounts for 20% of your final grade.	
Enrolment / Application	After receiving their Placement Test results students can enrol via Brightspace by first registering for the course and then selecting the group of their choice.	
Remarks	More information can be found on Brightspace WM1135TU	
Elective	Yes	
Tags	English proficiency	
Targetgroup	BSc, MSc and PhD students are welcome	
Category	BSc and MSc level	

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